Fragmented Governance, Density, and Housing Affordability: Evidence from Chicago's Aldermanic Privilege

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Abstract

Local land use regulation restricts housing supply and raises prices, but most research focuses on variation across municipalities. This paper examines how fragmentation of land use authority within cities affects housing markets, studying Chicago's "aldermanic privilege": an informal institution granting each of 50 aldermen effective veto power over development in their ward. I construct aldermanic "strictness scores" from permit processing times as a proxy for ward-level regulatory stringency and employ two empirical strategies: a spatial discontinuity design comparing density at ward boundaries, and event studies exploiting ward redistricting that quasi-randomly reassigned parcels to different aldermen. New construction is 10-16% less dense on the stricter side of ward boundaries. Rents rise 3–5% when blocks are reassigned to stricter aldermen, with home prices rising approximately 2%. These findings demonstrate that within-city fragmentation creates coordination failures: individual aldermen rationally restrict development, but the collective outcome is undersupply of housing relative to the social optimum.

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1 Introduction

Housing Prices are skyrocketing in major US cities and across the globe. Given the scarcity of vacant lots in dense urban areas, the primary way to add more housing supply is to construct taller and denser buildings that allow more people to live on the same-sized parcel of land. Density not only makes housing more affordable, but there are well-documented positive agglomeration spillovers from density, implying indirect benefits on top of the direct benefits of lower housing prices (Rossi-Hansberg et al., 2010; Ahlfeldt and Pietrostefani, 2019; Baum-Snow et al., 2024).

However, recent research in economics and other fields has highlighted that local land-use regulations, which raise the cost of building denser housing, make housing more expensive and drive people out of neighborhoods they have lived in for generations (Duranton and Puga, 2023).

Due to the ubiquitous nature of these regulations, this research has explored how these regulations come to be and the political economy surrounding them. A large strand of literature, pioneered by Fischel (2009), discussed how local homeowners have incentives to lobby for restrictive zoning policies to protect their property values. Additional research building on this idea has explored how local politics and the fragmentation of decision-making across municipalities can lead to under-provision of housing and other public goods with positive externalities (Hilber and Robert-Nicoud, 2013; Ortalo-Magné and Prat, 2014; Mast, 2024; Bordeu, 2025).

While the majority of this research has focused on fragmentation across municipalities Kulka et al. (2022); Monarrez and Schönholzer (2023), there is very little research on fragmentation within municipalities themselves. This mechanism can be equally as important as fragmentation across municipalities, especially in large cities where local political representation is often broken down into small districts or wards.

In this paper, I study the effect of within-municipality political fragmentation on housing supply in the context of the City of Chicago. While many cities have fragmented decision-

making, no major city in the United States gives as much power to its city council members (Aldermen) as Chicago. Importantly, Chicago lacks a city charter that clearly lays out the powers and restrictions of various elected offices of the city, so much of the power of the Aldermen is derived from tradition and informal norms. This unique level of power is often called "Aldermanic Privilege," and it allows Aldermen to exert significant control over land-use decisions in their wards (Thale, 2005). This privilege is so strong, in fact, that aldermen have been referred to as "little mayors" ruling their Wards as fiefdoms (Einhorn, 1991).

While this power has existed for many decades, it only recently has come under scrutiny as Chicago has faced a housing affordability crisis similar to many other major cities, with the Department of Housing and Urban Development alleging that the power has been used to ensure affordable housing is not built in certain affluent wards (Chase, 2023). In response to these allegations, Mayor Lightfoot signed an executive order attempting to limit the power of aldermanic privilege as part of a larger ethics reform package (Mayor's Press Office, 2019), but in practice not much has changed in the years since (Cherone, 2025). As a result, the debate continues today as to whether this power should be curtailed, or whether aldermen are still the best suited to make land-use decisions in their wards.[1]

To my knowledge, only one other paper has studied the effect of aldermanic privilege on housing supply in the city. Khan (2021) uses FOIA-requested rezoning data to look at the intensive-margin of aldermanic privilege, examining a specific function of the office where aldermen can personally approve or deny requests for zoning modifications on a specific lot. He finds that within 250 feet of a ward boundary, rezonings are $\approx 10\%$ smaller in terms of floor-area ratio (FAR), suggesting that aldermen are less willing to approve large rezonings when the benefits spill over into another ward, since the congestion costs are still likely to be internalized by the alderman's constituents even though the benefits are diffuse.

I expand on the work of Khan (2021) by focusing on the extensive-margin effects of aldermanic privilege. By looking at the density of all new construction in Chicago between

[1]See this link, where each city council candidate was asked specifically about their stance on "aldermanic prerogative": https://news.wttw.com/elections/voters-guide/2023/races/city-council

2006 and 2025 and not just specific rezoned properties, I can get a more holistic view of the aggregate effects of aldermanic privilege on housing supply in the city. In addition, I want to use my empirical results to investigate the fiscal impacts of this fragmentation, as it is possible that the aggregate effects of this under-provision of housing could have significant implications for city finances through lost property tax revenue and other channels. Given the city's structural budget deficit and stagnant population growth, examining policy levers by which policymakers can increase housing supply and tax revenue is of utmost importance.

My empirical strategy proceeds in two stages. First, I construct an "aldermanic strictness score" for each alderman using publicly available building permits data. This score captures the bureaucratic frictions that each alderman imposes on the development process, measured by permit processing times for high-discretion permits after residualizing out ward characteristics and applying empirical Bayes shrinkage. The resulting scores reveal substantial variation across aldermen that is not simply a reflection of geographic location or neighborhood demographics.

Second, I use these strictness scores to examine how development density varies at ward boundaries. Using a spatial regression discontinuity design that compares parcels on opposite sides of ward boundaries within the same zoning classification, I find that new construction is approximately 50-60% less dense on the stricter side of ward boundaries. These visual discontinuities are large and statistically significant, though they likely reflect both the causal effect of aldermanic behavior and developer self-selection into more lenient wards.

To isolate the causal effect of aldermanic strictness, I employ a border-pair fixed effects design that compares new construction within 250 feet of ward boundaries, controlling for zoning classification and construction-year by border-pair fixed effects. I find that a one standard deviation increase in aldermanic strictness is associated with a 10% decrease in dwelling units per acre, a 13% decrease in floor-area ratio, and a 16% decrease in total units. These effects attenuate substantially at greater distances from ward boundaries and the results are robust to a battery of specification checks including bandwidth sensitivity, donut-

hole exclusions, alternative polynomial orders, and placebo boundary tests, as documented in Section 4.4.

To examine whether these supply restrictions pass through into housing affordability, I exploit the 2015 and 2023 ward redistrictings as natural experiments. When ward boundaries are redrawn for population balancing, some census blocks are reassigned to new aldermen while nearby blocks remain under the same representation. This creates a difference-in-differences comparison where the identifying assumption is that reassigned and non-reassigned blocks would have followed parallel trends absent the change in aldermanic representation. Crucially, this design circumvents the selection problem inherent in the cross-sectional analysis: developers cannot anticipate or sort in response to redistricting that has not yet occurred.

I find that census blocks reassigned to stricter aldermen experience approximately 3–5% higher rents and 2% higher home sale prices relative to nearby blocks that did not switch aldermen. These effects emerge gradually over the post-redistricting period, consistent with a supply restriction channel where reduced new construction slowly tightens the housing market. The price and rent results complement the density findings: stricter aldermen approve less housing (Section 4), which restricts supply and raises equilibrium housing costs (Section 5).

The rest of the paper is organized as follows: Section 2 describes the data and institutional background of aldermanic privilege in Chicago. Section 3 presents a toy model of fragmented land-use authority and derives testable predictions. Section 4 presents the empirical results on development density at ward boundaries, and Section 5 presents the results on home prices and rents following ward redistricting. Section 6 discusses the policy implications of these findings, and Section 7 concludes.

2 Data and Institutional Background

I construct a novel dataset of all new construction in Chicago linked to every alderman-pair episode since 2006 using various publicly-accessible datasets from the Cook County Assessor's Office. In order to measure the density and frequency of new construction, I first use data on Single and Multi-Family Improvement Characteristics which allows me to observe every "improvement" for each parcel in Cook County. I merge this with the "Parcel Universe" dataset which gives me precise location coordinates. This allows me to limit my sample to just parcels within the City of Chicago, and to assign each parcel to its respective ward using shapefiles from the City of Chicago's data portal.[2] I then collapse this data to the first date I observe the parcel and consider that to be the construction date of the property, and create a cross-section of properties constructed between 2006 and 2025. Since this data is limited to parcels that have at most 6 units of housing, I then use the Commercial Valuation dataset from Cook County to infer multifamily housing density for buildings with more than 6 units and merge this with my residential conversion dataset to get the universe of new construction in Chicago between 2006 and 2025.

Other data products I use are the census and American Community Survey (ACS) data to create my independent variable, homeownership rate, at the ward level, and to control for various ward-level demographic characteristics in my regressions. I use zoning code data from 2nd city zoning [3] to include zoning code fixed effects in my regressions as well to ensure that changes in density at political borders do not come from mechanical changes in zoning.

In Table 1, I present summary statistics for my sample of new residential construction. I present statistics for my full sample in column (1), and then for multifamily buildings in column (2). In total, there are 13,236 new residential buildings constructed in Chicago between 2006 and 2025, and 2,469 of them were multifamily buildings with 2 or more units.

The total new housing stock has an average of 6.5 dwelling units and 2.3 stories, with an

[2] https://data.cityofchicago.org/
[3] https://secondcityzoning.org/zones/

average Floor Area Ratio (FAR) of 1.22 and Dwelling Units Per Acre of 28.47. Multifamily buildings are denser on average, with 31.3 dwelling units, 2.8 stories, a FAR of 2.3, and 74.5 dwelling units per acre. They are similar distances to their closest ward boundary at a median distance of 930 feet overall and 887 feet for multifamily buildings. In addition, both have positive average "strictness scores" as defined in Section 4. While this could be evidence that my strictness scores are not capturing pro vs. anti-development sentiment, it is also possible that demand is substantially higher in wards with stricter alderman, which is why the community elects them in an attempt to preserve ward character. In order to disentangle the effect of aldermanic influence from demand-side factors and neighborhood trends, more sophisticated empirical strategies are needed, as I discuss in Sections 4 and 5.

3 Toy Model

This framework adapts the infrastructure under-provision model of Bordeu (2025) to the context of housing supply, building on Khan (2021) who studies the same Chicago institutional setting.

The core tension in this model is between concentrated local costs and diffuse citywide benefits. While new housing creates positive spillovers for the city as a whole in the form of lower prices, increased tax revenue, agglomeration effects, the costs of development (construction disruption, congestion, neighborhood change) are borne primarily by immediate neighbors in the same ward.

3.1 Setup

Consider a city with two wards, A and B, each controlled by an alderman who determines the housing supply in their ward, denoted h_A and h_B respectively. Total citywide housing is $H = h_A + h_B$. The welfare of each ward depends on the following components:

- **Local Benefits** $b \cdot h_g$: Direct benefits from housing in ward g, including property tax

revenue and local retail demand. These accrue entirely to the ward where construction occurs.

- **Citywide Spillovers** $\phi(H)$: Aggregate benefits from total housing supply, including improved affordability, labor market pooling, and agglomeration economies. We assume $\phi'(H) > 0$ and $\phi''(H) < 0$ (diminishing marginal returns). Importantly, these benefits are diffuse—they are shared across the entire city.

- **Local Costs** $C(h_g)$: Congestion and nuisance costs from development, including traffic, noise, and strain on local amenities. These costs are concentrated in the ward where construction occurs, with $C'(h_g) > 0$ and $C''(h_g) > 0$ (convex costs).

- **Spillover Share** $\lambda_g \in (0, 1)$: The fraction of citywide spillover benefits $\phi(H)$ that accrue to residents of ward g. Because each ward contains only a subset of city residents, $\lambda_A + \lambda_B = 1$ and each $\lambda_g < 1$.

3.2 The Decentralized Problem (The Alderman)

Under aldermanic privilege, the alderman in ward g controls housing supply h_g to maximize local welfare. Crucially, the alderman internalizes the full local costs but only their ward's share of the citywide benefits. Taking the other ward's housing h_{-g} as given, the alderman solves:

$$\max_{h_g \geq 0} \quad V_g(h_g; h_{-g}) = b \cdot h_g + \lambda_g \cdot \phi(h_g + h_{-g}) - C(h_g) \tag{1}$$

The first-order condition equates the alderman's perceived marginal benefit to the marginal cost:

$$\underbrace{b + \lambda_g \cdot \phi'(H)}_{\text{Perceived Marginal Benefit}} \quad = \quad \underbrace{C'(h_g)}_{\text{Marginal Cost}} \tag{2}$$

The alderman perceives only fraction λ_g of the marginal spillover benefit $\phi'(H)$ because the remainder accrues to the other ward's residents.

3.3 The Planner's Problem

A citywide social planner internalizes that benefits generated in ward A also help residents in ward B. The planner chooses housing in both wards to maximize total welfare:

$$\max_{h_A, h_B \geq 0} \quad W = b \cdot (h_A + h_B) + \phi(H) - C(h_A) - C(h_B) \tag{3}$$

The planner's first-order condition equates the *full* marginal benefit to the marginal cost:

$$\underbrace{b + \phi'(H)}_{\text{True Marginal Benefit}} \quad = \quad \underbrace{C'(h_g)}_{\text{Marginal Cost}} \tag{4}$$

3.4 The Under-Provision Result

Comparing equations (2) and (4) reveals the structural inefficiency of decentralized control. Because $\lambda_g < 1$, the alderman perceives a lower marginal benefit than the planner:

$$b + \lambda_g \cdot \phi'(H) < b + \phi'(H) \tag{5}$$

Since the benefit function exhibits diminishing returns and costs are convex, the alderman's optimum occurs at a lower level of housing supply:

$$h_g^{\text{Alderman}} < h_g^{\text{Planner}} \tag{6}$$

The intuition is straightforward: the alderman stops approving housing once the local costs outweigh the *locally-captured* benefits, even though additional housing would generate positive spillovers to the rest of the city. The gap $(1 - \lambda_g)$ represents the share of benefits left on the table due to political fragmentation.

3.5 Spatial Variation: Why Borders Matter

The spillover share λ_g is not constant within a ward and varies by location. For housing built near the center of a ward, most of the neighborhood amenities and local spillovers accrue to fellow ward residents, so λ_g is relatively high. For housing built near the ward boundary, a larger share of spillovers (e.g., improved retail options, reduced congestion on shared roads) accrue to residents of the adjacent ward, so λ_g is lower.

Since the alderman's optimal housing supply is increasing in λ_g, this generates a spatial prediction within wards:

$$\frac{\partial h_g^*}{\partial \lambda_g} > 0 \quad \Longrightarrow \quad \text{Density is lower near ward boundaries} \tag{7}$$

This prediction is consistent with the regression discontinuity evidence in Figure 3, which shows a sharp drop in development density when moving from the lenient to the strict side of ward boundaries.

3.6 Heterogeneous Aldermanic Frictions

Aldermen may differ in the regulatory frictions they impose on the development process. Let $\kappa_g \geq 0$ represent alderman-specific costs of approving development—including bureaucratic delays, political costs from anti-development constituents, or personal preferences over neighborhood character. The alderman's problem becomes:

$$\max_{h_g \geq 0} \quad V_g = b \cdot h_g + \lambda_g \cdot \phi(H) - C(h_g) - \kappa_g \cdot h_g \tag{8}$$

The first-order condition is now:

$$b + \lambda_g \cdot \phi'(H) = C'(h_g) + \kappa_g \tag{9}$$

Higher κ_g raises the effective marginal cost of development, reducing equilibrium housing:

$$\frac{\partial h_g^*}{\partial \kappa_g} < 0 \tag{10}$$

Empirically, I construct a "strictness score" for each alderman based on permit processing times for high-discretion permits (Section 4). In the context of this model, the strictness score captures cross-alderman variation in κ_g. The border-pair fixed effects estimates in Table 2 test prediction (10) by comparing density on opposite sides of ward boundaries, finding that a one standard deviation increase in aldermanic strictness is associated with 10–16% lower density in new construction.

3.7 Implications for Housing Prices

If housing supply is restricted, standard urban economics predicts that equilibrium prices rise. Suppose housing demand in ward g is characterized by an inverse demand function $p_g = P(h_g)$ with $P'(h) < 0$. From (10), higher aldermanic frictions reduce housing supply. Combined with downward-sloping demand:

$$\frac{\partial p_g}{\partial \kappa_g} = P'(h_g) \cdot \frac{\partial h_g}{\partial \kappa_g} > 0 \tag{11}$$

Stricter aldermen restrict supply, which raises equilibrium prices. The event study in Section 5 tests this prediction by exploiting ward redistricting as a natural experiment. Census blocks reassigned to stricter aldermen experience approximately 4% higher home prices relative to nearby blocks that did not switch aldermen, consistent with the supply restriction channel.

3.8 Discussion

This framework generates four testable predictions that map directly to the empirical analysis:

1. **Under-provision** (eq. 6): Decentralized control leads to less housing than the social optimum.

2. **Border effects** (eq. 7): Density is lower near ward boundaries where spillovers to other wards are largest.

3. **Strictness effects** (eq. 10): Aldermen with higher regulatory frictions approve less housing.

4. **Price effects** (eq. 11): Supply restrictions from stricter aldermen raise housing prices.

The model clarifies the interpretation of the different empirical designs. The spatial RD estimates in Section 4 capture the total equilibrium difference in density across ward boundaries, which reflects both aldermanic approval decisions and developer sorting. The border-pair fixed effects estimates in Section 4 isolate the effect of aldermanic strictness conditional on location by comparing parcels within narrow bands of the same boundary. The redistricting event study in Section 5 provides the cleanest identification of the price effects, as parcels cannot select their alderman when boundaries are redrawn for population balancing.

4 Effects of Aldermanic Privilege on Density

4.1 Measuring Aldermanic Frictions

In order to examine the effects of aldermanic privilege I need a way to measure how much each alderman uses their privilege to influence development in their ward. I use publicly available

building permits data to construct this measure by measuring how long average processing times for *high-discretion* permits only. High-discretion permits are those designated as "New Construction, Renovation, Demolition, Porch Construction[4]" permits and permits that were reinstated after being previously revoked. I have excluded permits in the easy permit process and express permit program, along with signs and scaffolding permits. The goal is to target the specific permit types that tend to draw community opposition and are more likely to face delays due to community meetings and aldermanic discretion on behalf of their constituents.

Once I have my sample of permits, I construct the "aldermanic strictness score" as follows. First, I attempt to residualize out ward-specific characteristics by regressing the raw permit processing times on ward-level fundamentals such as median incomes, demographic characteristics, distance to the CBD and various other amenities, and month fixed-effects (Table A.1 in Appendix A reports the first-stage regression). I then estimate aldermanic fixed effects on these residualized processing times. Finally, since these estimates are a noisy signal of the "true" bureaucratic frictions caused by each alderman, I use empirical Bayes methods (Kline et al., 2022, 2024) to shrink these estimates towards zero based on how noisy each one is. Formally, the "two-stage" regression I run is:

$$\text{Processing Time}_i = \alpha + \mathbf{X}_w\beta + \gamma_m + \epsilon_i \tag{12}$$

$$\hat{\epsilon}_i = \mu + \delta_a + \eta_i \tag{13}$$

Where Processing Time$_i$ is the processing time of permit i, $\mathbf{X}_w$ is a vector of ward-level characteristics for the ward w that permit i is located in, and γ_m are month fixed effects. γ_a are the aldermanic fixed effects that I use as my dependent variable in my regressions. The scores are standardized to have mean zero and standard deviation one, and the results of my regressions can be found in Figure 1. In addition, a map of the aldermanic strictness scores can be found in Figure 2. There are strict and lenient aldermen all over the city, confirming I

[4]While these may seem minor, they have received extra scrutiny since the 2003 balcony collapse.

am not just picking up geographic patterns or neighborhood characteristics, but am actually capturing differences in the bureaucratic frictions created by each alderman. In the context of the model in Section 3, I interpret aldermen who have high strictness scores as having a lower λ (local capture parameter) leading to lower housing supply in their wards.

4.2 Visual Evidence of Density Discontinuities at Ward Borders

Before moving to my formal causal results using border-pair fixed effects, I first present some visual evidence of discontinuities in development density at ward borders. I employ a regression discontinuity design with a linear control function and triangular weights to estimate the gap in development density of new construction at the ward cutoff. I stack all ward borders together to have enough power to get precise estimates by placing wards with more lenient aldermen on the left side of the border (negative distance) and wards with stricter aldermen on the right side of the border (positive distance). Importantly, I restrict the sample to parcels that share the same zoning classification on both sides of the border, ensuring that differences in density are not mechanically driven by differences in allowable FAR across zoning districts. I use the rdrobust package Calonico et al. (2015) to implement this design with bias-corrected point estimates and robust standard errors.

In Figure 3 I present regression discontinuity plots for my two primary measures of density, DUPAC and FAR. I choose a bandwidth of 500 feet on either side of the border to focus on parcels that are the most likely to be similar in terms of neighborhood characteristics and local demand shocks, while still having enough observations to get precise estimates. In order to make them comparable, I plot the log of each density measure and estimate the discontinuity at the border. I find large and statistically significant drops in both measures of density when moving from the lenient side of the border to the strict side. Using the bias-corrected estimates from rdrobust, I estimate a 50% decrease in FAR and a 29% decrease in DUPAC when moving from the lenient side of the border to the strict side of a ward border.

The key assumption, as in any RD design, is that nothing else changes at the border

except for the alderman that the developer has to work with. If I were to claim causality from this design alone, I would need to claim that parcels cannot manipulate which side of the ward border they are on, but clearly developers can choose which ward to build in when considering development projects. This explains the large magnitude of the effects, as developers likely self-select into building in the more lenient wards if all other characteristics are equal between the two sides of the border. Regardless, these results are still suggestive that aldermanic behavior is affecting development density in their wards and potentially driving down aggregate densities in the city.

4.3 Do Stricter Aldermen Reduce Development Density?

Armed with these "strictness scores" and the visual motivating evidence from Figure 3, I turn now to my main empirical question: do stricter aldermen reduce development density in their wards?

To answer this question I run regressions of the following form:

$$\ln(Y_{iwbt}) = \beta \cdot \text{Strictness}_w + \mathbf{X}'_w\Theta + \alpha_z + \mu_b + \gamma_t + \varepsilon_{iwbt} \tag{14}$$

where Y_{iwbt} represents the density outcome for parcel i located in ward w along border-pair b, constructed in year t.

My three density outcomes of interest are common in the literature: Dwelling Units Per Acre (DUPAC), which is defined as the number of dwelling units on the parcel divided by the parcel's acreage; Floor Area Ratio (FAR), defined as the total building square footage on the parcel divided by the parcel's land area; and Units in the building.

The variable of interest is Strictness_w, which denotes the standardized bureaucratic friction score for the alderman of ward w. The vector $\mathbf{X}_w$ includes time-varying ward-level controls, including median household income, racial composition, and education levels.

Critically, I include a stringent set of fixed effects to isolate the causal effect of aldermanic

strictness on development density. α_z are zoning classification fixed effects, which ensure that comparisons are only made within the same zoning classification on each side of the border. For example, if one side of the border is zoned for RT-3.5, which allows a maximum FAR of 1.05, and the other side is zoned for RM-5.5, which allows for a maximum FAR of 2.5, there will be a mechanical difference in densities at the border that is not the result of aldermanic influence.

In addition, following Black (1999); Bayer et al. (2007); Kulka et al. (2022), I include border-pair fixed effects μ_b, which are interacted with zoning code fixed effects to ensure comparisons are being made within border-pairs and within zoning classifications. I also include construction year fixed effects γ_t to absorb city-wide trends in development density over time. While this is very strict and limits the amount of variation in my sample, differential trends in development density across different parts of the city (i.e. the West Loop boom and Pilsen gentrification) could otherwise confound my estimates.

And finally, I limit the sample to parcels with between 2 and 100 units to focus on mid-size multifamily housing. Single family housing is the vast majority of new construction and does not have significant differences across political borders, and large developments (100+ units) are relatively rare and typically go through a more complicated planned development (PD) process with more oversight from the city and alderman.

The main results are presented in Table 2. I limit the sample to parcels within 250 feet of a ward border to specifically test if different aldermen affect density at a very fine scale where local demand shocks and neighborhood characteristics are likely to be very similar on either side of the border. The results are quite substantial and consistent across all three density measures. A one standard deviation increase in my aldermanic strictness measure is associated with a 10% decrease in DUPAC, a 13% decrease in FAR, and a 16% decrease in the number of units in the building, among new construction built in the same year and zoned the same on either side of the border.

In addition, to show that these differences are not driven by area characteristics and

appear to be hyper-localized to the border where aldermanic differences are likely to have the most effect, I also run the same regressions but limit the sample to parcels within 1000 feet of a ward border. There is still some effect of aldermanic strictness on the number of units, but the magnitude is substantially smaller. The other coefficients are still negative but much smaller and not significantly different from zero, as shown in Table B.1 in the Appendix. As a result, I interpret these results as strong evidence that aldermanic privilege and the bureaucratic frictions they create have a significant effect on development density in their wards.

To test whether aldermanic strictness is correlated with smoothly varying neighborhood characteristics at the ward border, I regress block group-level demographics on the strictness score using the same fixed effects specification. I assign each parcel to a census block group via spatial intersection with 2019 American Community Survey geographies and merge demographics from the corresponding block group. Table B.2 in the Appendix presents balance tests for eight demographic variables: racial composition, median income, education levels, homeownership rates, median rent, home values, and population density. I find null results across all variables, suggesting that the differences in density at ward borders are driven by aldermanic behavior rather than demographic sorting.

4.4 Robustness of the Spatial Discontinuity

The spatial regression discontinuity results presented in Section 4.2 are subject to several potential concerns common to RD designs. In this section, I present bandwidth sensitivity in the main text and provide additional robustness checks in Appendix B that address concerns about observations precisely at the boundary, functional form assumptions, and the validity of the discontinuity itself.

4.4.1 Bandwidth Sensitivity

A key choice in any RD design is the bandwidth, which is the window around the cutoff within which observations are used for estimation. My main specification uses a 500-foot bandwidth, but results should be robust to reasonable alternatives. Figure 4 presents RD plots for both log(DUPAC) and log(FAR) at bandwidths of 250, 500, and 1,000 feet. The discontinuity remains visible and statistically significant across all bandwidth choices for both density measures, though precision naturally decreases with narrower bandwidths due to smaller sample sizes. The point estimates are relatively stable, suggesting the results are not driven by a particular bandwidth choice.

4.4.2 Additional Robustness Checks

Additional robustness checks are presented in Appendix B. These include: (1) a "donut RD" (Barreca et al., 2011, 2016) specification that excludes parcels within 25 or 50 feet of the ward boundary to address concerns about observations precisely at the boundary (Figure B.1); (2) alternative functional forms using quadratic and cubic polynomial specifications (Figure B.2); and (3) placebo boundary tests at artificial cutoffs 500 feet inside each ward (Figure B.3). All robustness checks support the main findings.

Taken together, these robustness checks provide strong evidence that the density discontinuity at ward boundaries is a genuine phenomenon rather than an artifact of bandwidth selection, boundary-specific idiosyncrasies, functional form assumptions, or spurious spatial patterns. Combined with the covariate balance tests in Table B.2 showing no discontinuity in demographic characteristics, these results support the interpretation that aldermanic behavior causally affects development density near ward boundaries.

5 Event Study Evidence: Rents and Home Prices

5.1 Event Study Design

The cross-sectional evidence in Section 4 establishes that new construction is less dense on the stricter side of ward boundaries. While density on its own is an important outcome, policymakers tend to be more interested in its implications for housing affordability. To look at downstream effects on rents and home values, I exploit the 2015 and 2023 ward redistrictings as natural experiments to examine how housing costs respond when parcels are reassigned to aldermen with different levels of strictness.

5.2 Identification Strategy and Empirical Specification

Ward boundaries in Chicago are redrawn following each decennial census to equalize populations across the city's 50 wards. The 2015 remap (based on the 2010 Census) and the 2023 remap (based on the 2020 Census) each reassigned many parcels to new aldermen while leaving nearby parcels under the same ward. This creates a difference-in-differences style comparison: census blocks that switched aldermen (treated) versus nearby blocks that remained under the same alderman (control). The identifying assumption is that, absent the change in aldermanic representation, treated and control blocks would have followed parallel trends in housing costs.

This design exploits the fact that ward boundaries shift for reasons unrelated to housing market conditions in specific census blocks. Redistricting in Chicago occurs every ten years following the decennial census, with boundaries redrawn to equalize population across wards. The resulting boundary changes create quasi-random variation in alderman assignment for blocks near the old and new boundaries.

Control Group Construction. A treated block that switches from Ward A to Ward B could potentially be compared to control blocks that remained in either ward. However,

blocks that remained in Ward B were never subject to Ward A's alderman, making them a poor counterfactual for how the treated block would have evolved absent redistricting. I therefore restrict the control group to blocks that remained in the *origin* ward—that is, blocks near the same ward boundary that did not switch and thus experienced the same pre-treatment aldermanic policies as the treated blocks. This ensures that treated and control blocks share a common policy environment in the pre-period. To avoid contamination from electoral turnover, I also exclude control blocks in wards where the alderman changed via election (rather than redistricting) during the sample window.

Figure 5 illustrates this design with a concrete example for Wards 13 and 23 during the 2015 redistricting. The top panel shows ward assignments before redistricting, the middle panel shows assignments after, and the bottom panel shows the resulting treatment status based on the change in alderman strictness. Blocks that switched from Ward 13 to Ward 23 are compared only to control blocks that remained in Ward 13, not to blocks that remained in Ward 23. Figure 6 shows the citywide distribution of treatment and control blocks for the 2015 and 2023 redistricting events.

Empirical Specification. I estimate the following event study specification at the listing level (for rentals) and transaction level (for sales):

$$y_{it} = \alpha_{ps} + \lambda_{pt} + X_{it}'\gamma + \sum_{\tau=-5,\tau\neq-1}^{T} \beta_\tau \left(\Delta\text{Strictness}_i \times \mathbf{1}[t=\tau]\right) + \varepsilon_{it} \tag{15}$$

where y_{it} is the log rent or log sale price for listing (or transaction) i at event time t. $\Delta\text{Strictness}_i$ is the change in alderman strictness for the block containing the observation, defined as the strictness score of the destination alderman minus the origin alderman. To ensure treatment intensity is predetermined, strictness scores are measured in the year prior to redistricting (2014 for the 2015 redistricting cohort; 2022 for the 2023 cohort). X_{it} is a vector of property-level hedonic controls.

The key fixed effects ensure that comparisons are made between nearby observations

experiencing the same local shocks. α_{ps} is a border-pair-by-origin-side fixed effect, ensuring that treated and control observations (i) are near the same ward boundary and (ii) were in the same origin ward prior to redistricting. λ_{pt} is a border-pair-by-year fixed effect, which allows for flexible time-varying shocks at each ward boundary—a stronger control than simple year fixed effects, as it absorbs any local trends (e.g., gentrification, new transit) that affect both sides of a given boundary. For the stacked rental specification, these fixed effects are interacted with cohort indicators. Standard errors are clustered at the census block level to account for serial correlation within geographic units.

The coefficients β_τ trace out the effect of aldermanic strictness on housing costs in event time, with $\tau = -1$ (the year before redistricting takes effect) as the omitted reference period. The identifying assumption implies $\beta_\tau = 0$ for $\tau < 0$: prior to redistricting, there should be no systematic relationship between future changes in alderman strictness and housing cost trends within the same border-pair-side cell.

I restrict the sample to observations within 1,000 feet of ward boundaries and apply triangular kernel weights that give more weight to observations closer to the boundary, where treated and control units are most comparable. In the continuous treatment specification, ΔStrictness enters as a continuous variable; I also present binary specifications that allow for separate effects by treatment direction (moved to stricter versus moved to more lenient).

Predictions. If stricter aldermen constrain housing supply, standard urban economics predicts that both rents and home prices should rise when blocks are reassigned to stricter aldermen. The event study design allows me to trace out the dynamics of this response and verify that effects emerge only after redistricting, supporting a causal interpretation.

Data and Sample. For rental listings, I use scraped listing data from 2014–2025, which allows me to stack both the 2015 and 2023 redistricting events. This stacked design maximizes statistical power by pooling two natural experiments. The rental sample includes approximately 3.9 million listings within 1,000 feet of ward boundaries. For home sales, I

use transaction data from the Cook County Recorder of Deeds for the 2015 redistricting cohort, with a sample window from 2010–2020. The home sales sample includes approximately 27,000 transactions near ward boundaries.[5]

Table 3 presents summary statistics for the event study samples. Treated and control groups are broadly similar on observable property characteristics: unit sizes and building type distributions are comparable across treatment groups for rentals, as are mean building size, age, and bedroom counts for home sales. This balance supports the identifying assumption that redistricting-induced changes in alderman assignment are orthogonal to pre-existing neighborhood characteristics.

5.3 Results: Rents

Figure 7 presents the main event study results for rental listings, stacking the 2015 and 2023 redistricting cohorts. Panel A shows the continuous treatment specification, where treatment intensity is measured by the change in alderman strictness. The coefficients trace out the effect of moving to an alderman who is one standard deviation stricter, relative to rents in the year before redistricting takes effect.

The pre-period coefficients are stable around zero, supporting the parallel trends assumption. Following redistricting, rents in areas assigned to stricter aldermen diverge from the reference period. By two years after redistricting, moving to an alderman one standard deviation stricter is associated with approximately 3–5% higher rents. This effect is persistent and grows slightly over time, consistent with a supply restriction channel where reduced new construction slowly tightens the rental market.

Panel B presents the binary specification, which relaxes the linearity assumption and allows for separate effects by treatment direction. Here, treatment is defined as switching to an alderman with a higher strictness score ("moved to stricter") or the reverse ("moved to more lenient"), with the control group consisting of census blocks that did not switch

[5] I focus on the 2015 redistricting for home sales because the 2023 redistricting provides limited post-period data. Appendix C presents additional robustness checks.

aldermen. The results are striking: blocks that moved to stricter aldermen experience rising rents, while blocks that moved to more lenient aldermen see rents *fall* by approximately 5–7%. This symmetric pattern—with effects in opposite directions matching theoretical predictions—provides strong evidence against the results being driven by specification search or unobserved confounds. It is difficult to p-hack one's way into two effects with opposite signs that both align with theory.

Interpretation. The rental results provide clean evidence that aldermanic strictness affects housing costs. The flat pre-trends, sharp break at redistricting, and symmetric effects by treatment direction all support a causal interpretation. The magnitude is economically meaningful: a 3–5% increase in rents corresponds to approximately $40–70 per month for the median rental unit in my sample. Combined with the density results from Section 4, these findings suggest that aldermanic privilege has quantitatively important effects on both housing supply and affordability.

5.4 Results: Home Sale Prices

Figure 8 presents event study results for home sale prices using the 2015 redistricting. Panel A shows the continuous treatment specification. The pattern is qualitatively consistent with the rental results, though noisier due to the smaller sample size and greater volatility of home prices relative to rents.

The pre-period coefficients show some variation but are generally centered around zero. Following redistricting, home prices in areas assigned to stricter aldermen rise relative to the reference period, reaching approximately 2% higher by year 5. The post-period trajectory shows a gradual increase rather than a sharp level shift, which may reflect the slower capitalization of expected future supply restrictions into asset prices.

Panel B presents the binary specification for home sales. Despite larger confidence intervals, the pattern is directionally consistent with theory: blocks that moved to stricter

aldermen show positive price effects, while blocks that moved to more lenient aldermen show slight declines. The effects are less precisely estimated than for rents, and I cannot reject that the two groups have equal price trajectories at conventional significance levels.

Interpretation. The home sales results are more suggestive than definitive, reflecting the inherent challenges of studying asset prices: smaller sample sizes, greater price volatility, and the difficulty of isolating supply-side effects from demand-side factors that may correlate with aldermanic strictness. Nevertheless, the directional consistency with the rental results—and with the theoretical prediction that supply restrictions raise prices—provides supporting evidence that aldermanic privilege affects housing costs across both the rental and owner-occupied segments of the market.

The magnitude of the home price effect, while imprecisely estimated, is economically plausible. A 2% increase in home prices corresponds to approximately $5,000 for the median home in my sample, representing a transfer from buyers to incumbent homeowners.

5.5 Pooled Difference-in-Differences Estimates

Table 4 presents pooled difference-in-differences estimates that collapse the post-period effects into a single coefficient. The pooled estimates confirm the event study patterns: moving to a stricter alderman is associated with 3–4% higher rents and approximately 2% higher home prices. Both panels include specifications with and without property-level hedonic controls. Point estimates are somewhat attenuated with hedonic controls, consistent with part of the effect operating through changes in housing stock composition induced by aldermanic supply restrictions—but the results remain positive and economically meaningful even after controlling for observable property characteristics.

6 Discussion

6.1 Interpretation of Estimates

It is useful to distinguish three margins along which aldermanic strictness can affect housing supply. First, the intensive margin: conditional on a developer seeking discretionary approval (such as a rezoning or planned development designation), how does aldermanic strictness affect the size and density of the approved project? This is the margin studied by Khan (2021), who finds that rezonings are approximately 10% smaller near ward boundaries where benefits spill over to adjacent wards.

Second, the extensive margin of realized development: how does aldermanic strictness affect what actually gets built, including projects that proceed by-right without seeking discretionary approval? This is the margin I study. By examining all new construction and not just rezoned parcels, I capture an additional channel: developers who respond to strict aldermen by building smaller by-right projects that do not require aldermanic sign-off, rather than proposing larger projects that would trigger discretionary review. My density estimates thus reflect both the intensive-margin effect documented by Khan (stricter aldermen approve smaller projects) and an extensive-margin effect (developers in strict wards avoid discretionary review entirely by building smaller).

Third, there is the deterrence margin: projects that are never proposed because developers anticipate rejection, face prohibitive uncertainty, or choose to invest in other jurisdictions entirely. This margin is by definition unobserved in permit or construction data. If developers avoid strict wards at the site-selection stage, before any permit application is filed, then even my estimates of realized construction understate the true effect of aldermanic privilege on housing supply.

The estimates presented in this paper capture the first two margins but not the third. They are therefore best interpreted as lower bounds of the total effect of aldermanic privilege on housing supply. Several mechanisms contribute to the unmeasured extensive margin of

deterred development.

First, regulatory uncertainty deters risk-averse developers even when expected approval probability is reasonable. A developer considering a 50-unit apartment building may abandon the project if there is substantial probability of rejection, even if the most likely outcome is approval. The option value of waiting or building elsewhere may dominate proceeding with an uncertain approval process.

Second, the negotiation process itself imposes costs. Aldermen may approve projects conditional on design modifications, density reductions, or community benefit agreements that are not reflected in final permit data. The anticipation of these costs may deter marginal projects from being proposed in the first place, even if they would ultimately be approved in some modified form.

Third, the mere existence of aldermanic discretion creates a "chilling effect" on development proposals. Developers report that navigating aldermanic approval is among the most uncertain and time-consuming aspects of Chicago development, leading some to focus their investments in suburban jurisdictions with more predictable approval processes. This reallocation of development activity away from the city represents a real cost that does not appear in any permit database.

These unmeasured channels suggest that the welfare costs of aldermanic privilege extend beyond what my reduced-form estimates can capture. A complete accounting would require modeling developer entry and site selection decisions, which I hope to pursue in future work.

6.2 Connection to the Broader Literature

My findings contribute to several strands of the urban economics literature. Most directly, this paper adds to the extensive evidence that land use regulations restrict housing supply and raise prices (Glaeser et al., 2005; Saiz, 2010; Turner et al., 2014). The contribution here is to show that this mechanism operates at the hyper-local level within a single city, not only across metropolitan areas or municipalities with different levels of statuatory zoning regu-

lations. Ward-level variation in regulatory stringency creates meaningful supply distortions even within a unified municipal boundary.

The results also speak to the political economy of land use regulation. Fischel (2009) argues that homeowners have strong incentives to support restrictive zoning to protect their property values. My price results are consistent with this mechanism: rents rise 3–5% and home prices rise approximately 2% when blocks are reassigned to stricter aldermen. But this price appreciation represents a *transfer* from prospective buyers to incumbent homeowners, not a social welfare gain. Aldermanic privilege thus functions as a mechanism through which homeowners can extract rents via political channels, with costs borne by prospective residents, renters, and the city as a whole through reduced tax base and diminished agglomeration economies.

More broadly, this paper contributes to a growing literature on how the fragmentation of governance affects urban outcomes. Prior work has emphasized fragmentation across municipalities: Tiebout sorting, fiscal zoning, and the tendency for suburban jurisdictions to externalize the costs of housing restrictions onto regional housing markets (Hilber and Robert-Nicoud, 2013; Kulka et al., 2022; Monarrez and Schönholzer, 2023; Bordeu, 2025). My results demonstrate that fragmentation within cities can generate similar coordination failures. Chicago is nominally a unified municipality with centralized zoning authority, but the informal institution of aldermanic privilege effectively Balkanizes land use decision-making across 50 independent actors.

This fragmentation creates a classic collective action problem. Housing affordability is a citywide public good: the benefits of new construction such as lower prices, expanded tax base, and agglomeration spillovers are diffuse and accrue across the metropolitan area. But the costs of new development, which include construction disruption, traffic congestion, neighborhood change, are concentrated and borne primarily by residents of the ward where construction occurs. Each alderman, rationally maximizing the welfare of their constituents, will restrict development to a greater degree than a social planner who internalizes citywide

benefits. The result is systematic undersupply of housing relative to the social optimum.

This interpretation is consistent with Khan (2021), who finds that aldermen approve smaller rezonings near ward boundaries where benefits spill over to adjacent wards. My paper complements his work by documenting the aggregate consequences of this behavior along a broader margin: not just smaller individual projects conditional on seeking approval, but systematically lower realized density and higher prices in wards governed by stricter aldermen.

6.3 Policy Implications

The results of this paper inform ongoing debates about reforming aldermanic privilege in Chicago. Mayor Lightfoot's 2019 executive order attempted to limit aldermanic discretion over development approvals, and the U.S. Department of Housing and Urban Development has raised concerns that the institution has been used to block affordable housing in affluent wards (Chase, 2023). My findings suggest these concerns have empirical foundation: aldermanic strictness has quantifiable effects on both housing supply and prices.

Several policy approaches could address the coordination failures documented here. One option is to centralize land use authority, shifting discretion from individual aldermen to citywide bodies such as the Plan Commission or Zoning Board of Appeals. Centralized decision-makers would internalize the citywide benefits of housing production that individual aldermen rationally ignore. The tradeoff is that centralization sacrifices local knowledge—aldermen often have legitimate information about neighborhood conditions and constituent preferences that citywide bodies lack—and may reduce democratic accountability for land use decisions.

A second approach is to expand by-right development, reducing the scope of aldermanic discretion for projects that comply with existing zoning. If a proposed building meets all code requirements, approval would be automatic rather than subject to aldermanic review. This approach preserves the zoning code as the mechanism for expressing community preferences

about land use while eliminating the uncertainty and delay associated with discretionary review. The limitation is that by-right development reduces flexibility to address legitimate site-specific concerns that may not be anticipated by general zoning categories.

A third approach focuses on transparency, requiring aldermen to publicly state their positions on pending development applications and creating systematic records of approval and rejection decisions. Greater transparency could enable electoral accountability: voters who prioritize housing affordability could identify and oppose aldermen with restrictive track records. However, transparency alone may not change underlying political incentives if ward electorates are dominated by homeowners who benefit from restriction.

These approaches are not mutually exclusive, and the optimal policy likely involves some combination. The key insight from this paper is that the status quo of decentralized decision making is costly. Chicago's current institution of aldermanic privilege generates measurable distortions to housing supply and prices. Whether the benefits of local control—democratic accountability, local knowledge, responsiveness to neighborhood concerns outweigh these costs is ultimately a political judgment. But that judgment should be informed by evidence on the magnitude of the tradeoffs involved.

7 Conclusion

This paper examines how the fragmentation of land use authority within cities affects housing supply and prices. I study Chicago's "aldermanic privilege": the informal institution granting each of the city's 50 aldermen effective veto power over development in their ward—and document its consequences for housing markets.

I construct aldermanic strictness scores from permit processing times, using a residualization approach that removes the influence of ward characteristics and an empirical Bayes procedure that addresses measurement error for aldermen with few permits. The resulting scores reveal substantial and persistent variation in regulatory stringency across aldermen

that is not simply a reflection of geography or neighborhood demographics.

Using these scores, I employ two complementary empirical strategies. First, a spatial discontinuity design comparing development density on opposite sides of ward boundaries finds that new construction is 10-16% less dense on the stricter side of ward boundaries, with effects concentrated within 250 feet of the boundary. Second, event studies exploiting the 2015 and 2023 ward redistrictings show that rents rise 3–5% when census blocks are reassigned to stricter aldermen. Home prices rise approximately 2% following reassignment, though these estimates are noisier due to smaller sample sizes.

These findings have several implications. First, within-city fragmentation of land use authority can generate coordination failures analogous to those documented across municipalities: individual decision-makers rationally restrict development, but the collective outcome is undersupply of housing relative to the social optimum. Second, the effects of regulatory stringency are highly localized—concentrated near ward boundaries where the jurisdiction of strict and lenient aldermen meet—suggesting that developers are attentive to aldermanic preferences at fine spatial scales. Third, the symmetric effects in the binary specification—rents rising under stricter aldermen and falling under more lenient ones—provide strong internal validity for the supply restriction mechanism.

Several limitations warrant mention. The estimates presented here do not capture the extensive margin "deterrence" effects and likely represent lower bounds of the true impact, as the considered projects that are never proposed are unobserved. The analysis focuses on local, partial-equilibrium effects and does not capture citywide general-equilibrium adjustments to changes in aldermanic strictness. And while the event study design supports causal interpretation, I cannot fully separate supply-side mechanisms (aldermen restricting construction) from demand-side mechanisms (stricter aldermen signaling neighborhood desirability) in explaining price effects.

Future work could extend this analysis in several directions: quantifying the welfare costs of aldermanic privilege using a structural model of developer and household location choice;

examining whether aldermanic strictness affects the composition of housing supply, particularly affordable versus market-rate units; and studying whether electoral competition or transparency requirements discipline restrictive behavior. Extending the empirical approach to other cities with ward-based governance could assess the external validity of these findings.

Chicago's ongoing debate over aldermanic privilege is not merely symbolic. The institution has real, quantifiable effects on where housing gets built and how much it costs. As cities across the United States grapple with housing affordability crises, understanding how local political institutions shape housing supply is essential. The evidence presented here suggests that reforming fragmented land use authority through some combination of centralization, by-right development, or other mechanisms could meaningfully improve housing market outcomes.

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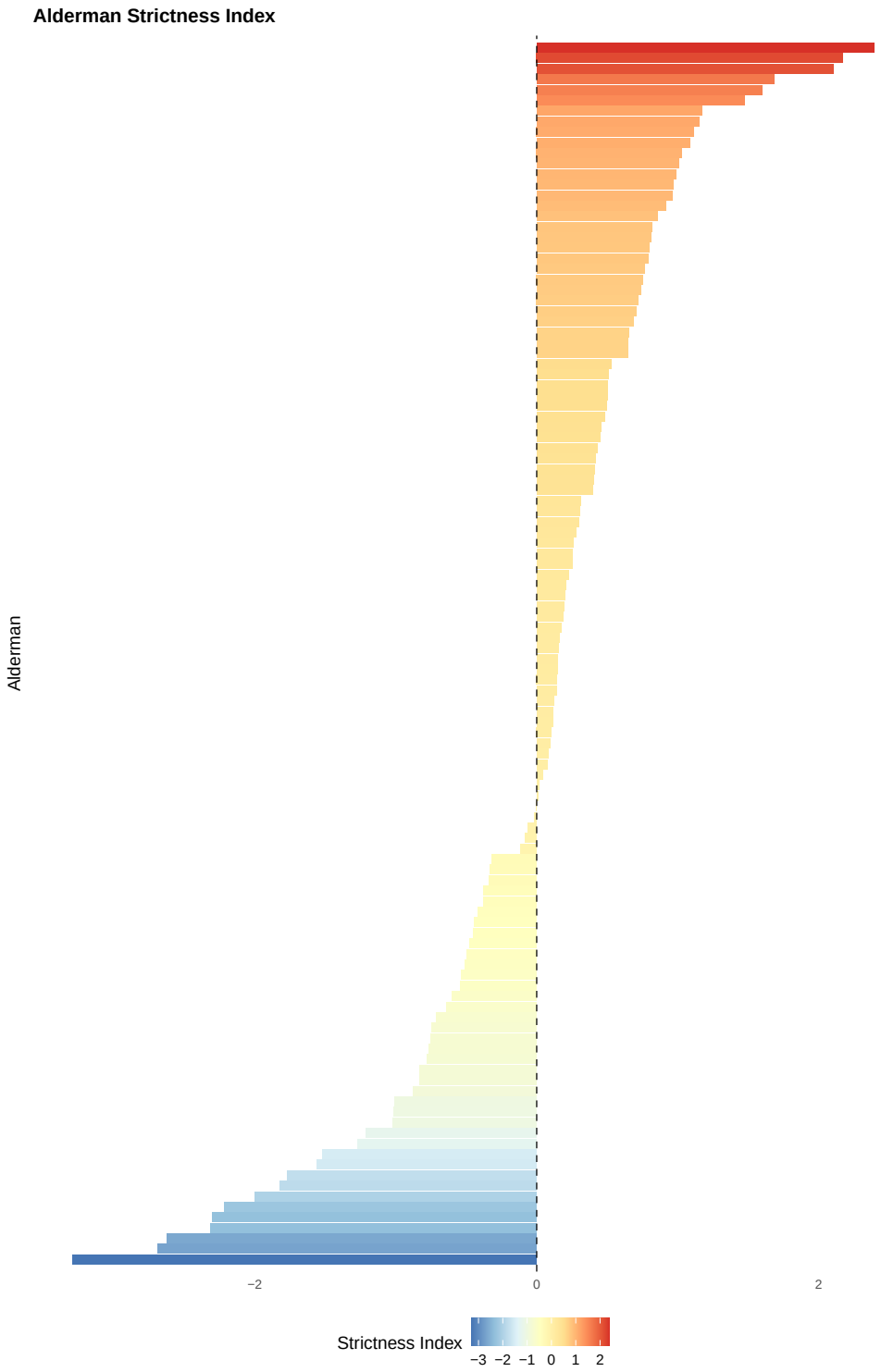


Figure 1: Distribution of Aldermanic Strictness Scores

Notes: Strictness scores are constructed from residualized permit processing times with empirical Bayes shrinkage. Scores are standardized to mean zero. Higher values indicate aldermen associated with longer permit processing times after controlling for ward characteristics and month fixed effects.

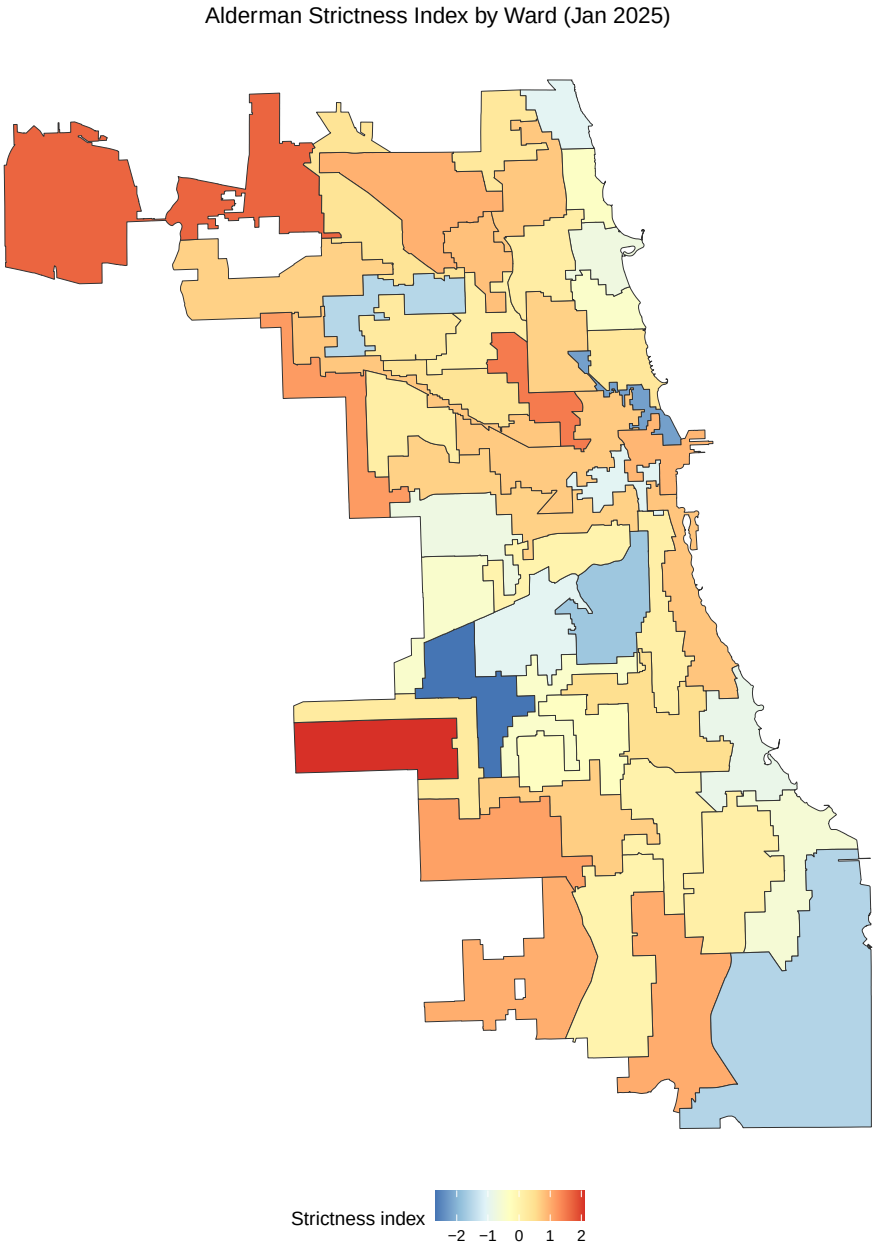


Figure 2: Geographic Distribution of Aldermanic Strictness

Notes: Map shows strictness scores by ward. Darker shades indicate stricter aldermen.

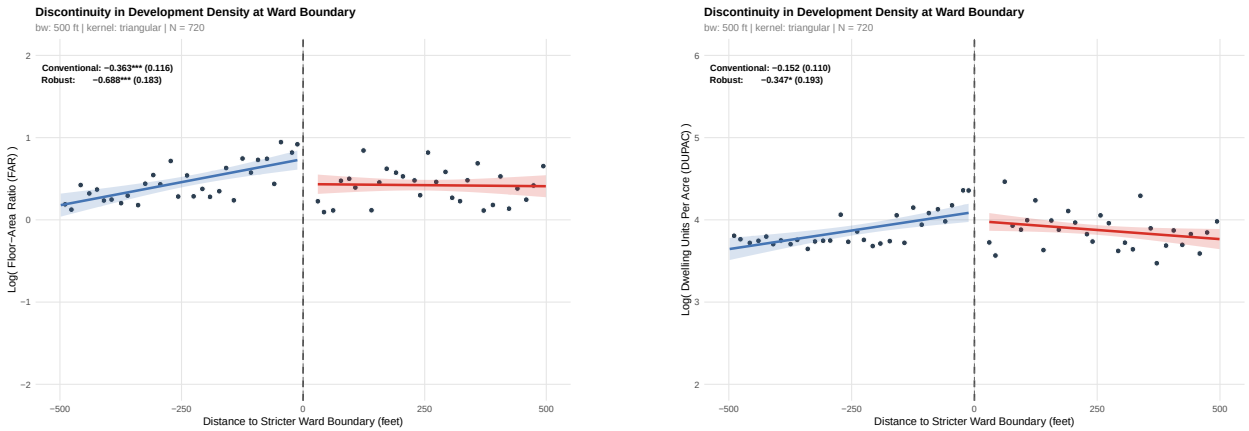


Figure 3: Regression Discontinuity at Ward Boundaries: Log(FAR) and Log(DUPAC)

Notes: Sample restricted to new multifamily construction (2–100 units) within 500 feet of ward boundaries where both sides share the same zoning classification. Negative distance indicates the more lenient alderman's side; positive distance indicates the stricter side. Local linear regression with triangular kernel weights. Bias-corrected point estimates and robust confidence intervals from rdrobust. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

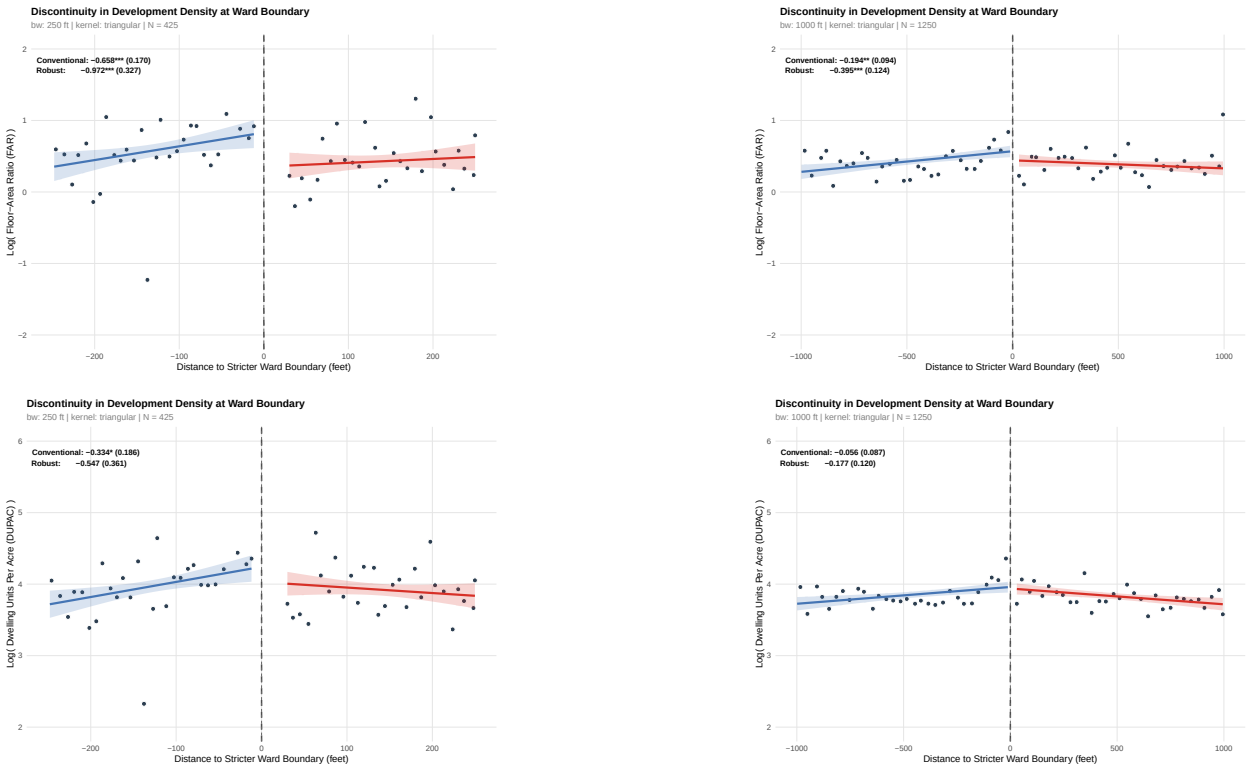


Figure 4: Bandwidth Sensitivity for RD Estimates

Notes: Top row: Log(FAR); bottom row: Log(DUPAC). Left column: 250-foot bandwidth; right column: 1,000-foot bandwidth. Main specification uses 500-foot bandwidth (Figure 3). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

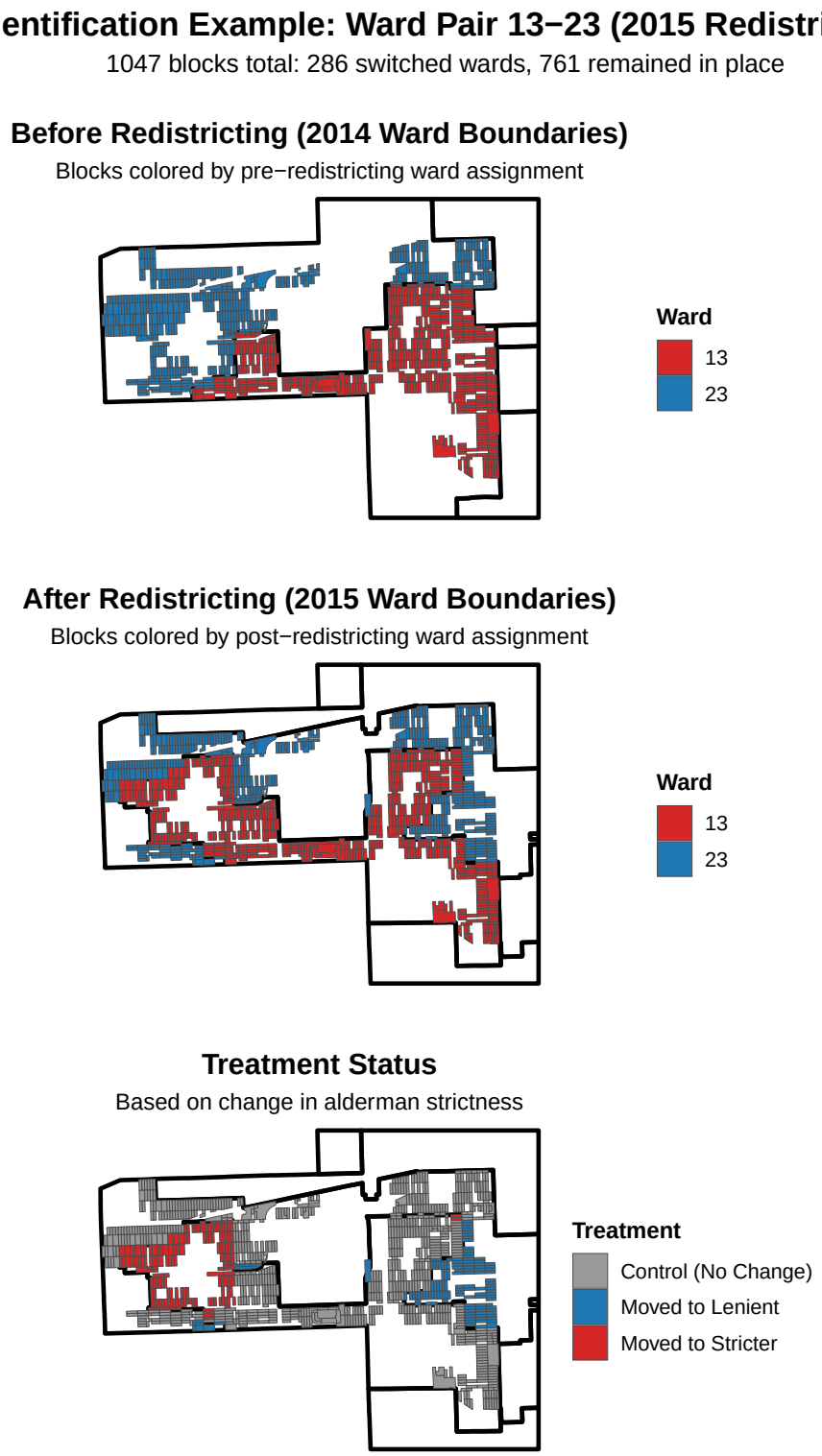


Figure 5: Identification Example: Ward Pair 13–23 (2015 Redistricting)

Notes: Top panel shows ward assignments before redistricting (2014 boundaries), middle panel shows assignments after redistricting (2015 boundaries), and bottom panel shows treatment status based on the change in alderman strictness. Blocks switching from Ward 13 to Ward 23 are compared to controls that remained in Ward 13 (the origin ward), and vice versa. Sample restricted to blocks within 1,000 feet of ward boundaries.

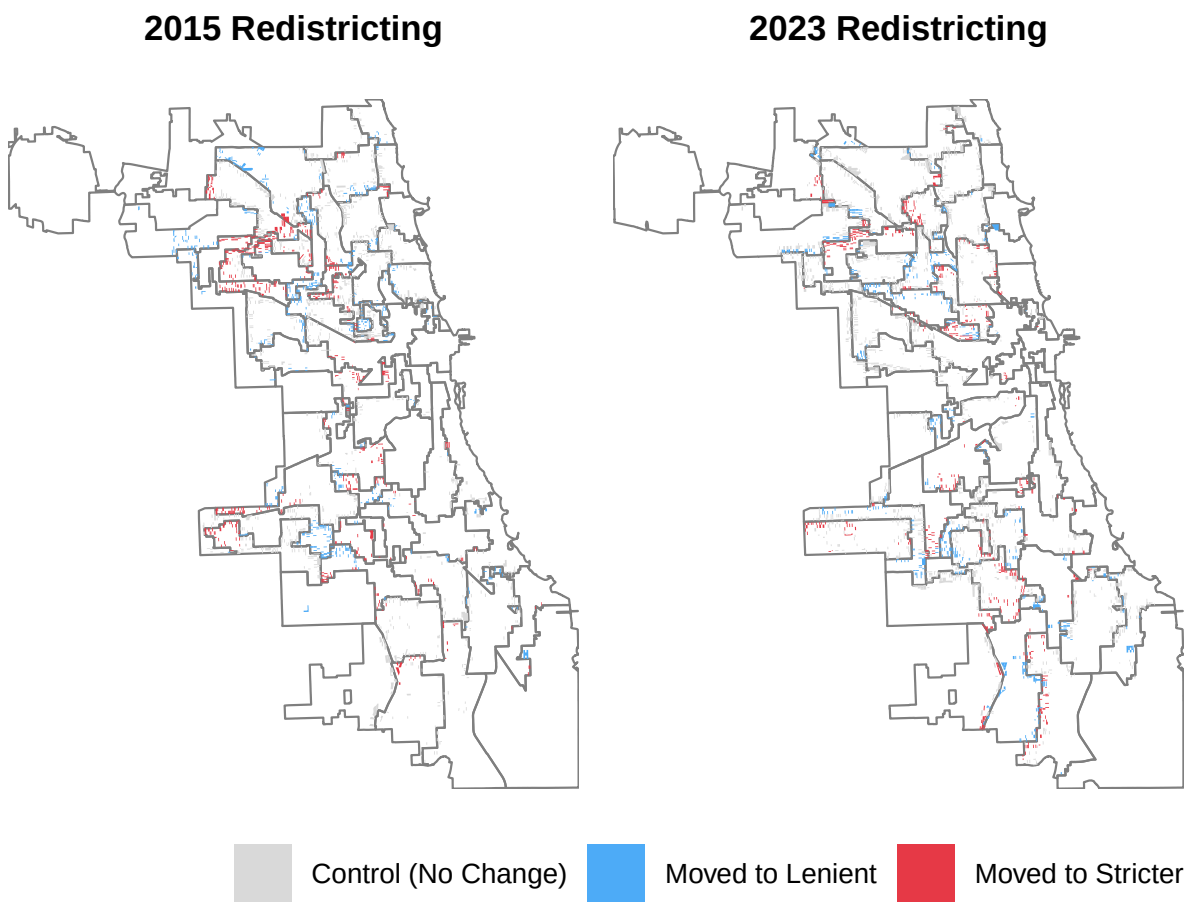
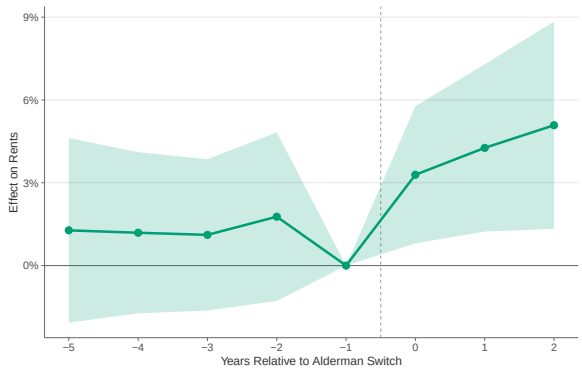


Figure 6: Treatment and Control Blocks: Citywide Distribution

Notes: Geographic distribution of treated and control blocks for the 2015 redistricting (left) and 2023 redistricting (right). Red indicates blocks that moved to a stricter alderman, blue indicates blocks that moved to a more lenient alderman, and gray indicates control blocks that did not switch wards. Sample restricted to blocks within 1,000 feet of ward boundaries. Controls exclude wards with aldermanic turnover via election.



Panel A: Continuous Treatment

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Panel B: Binary Treatment

Figure 7: Event Study: Effect of Alderman Strictness on Rents

Notes: Panel A: continuous treatment (effect of 1 SD increase in strictness). Panel B: binary treatment (separate effects for blocks moving to stricter vs. more lenient aldermen). Stacked estimator combining 2015 and 2023 redistricting cohorts. Cohort-by-border-pair-by-side and cohort-by-border-pair-by-year fixed effects. Triangular kernel weights with 1,000 ft bandwidth. Includes hedonic controls for building type, square footage, bedrooms, and bathrooms. Year -1 omitted as reference. Post-period truncated at $t = 2$ where both cohorts contribute observations. 95% confidence intervals based on standard errors clustered at the census block level.

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Panel A: Continuous Treatment

Panel B: Binary Treatment

Figure 8: Event Study: Effect of Alderman Strictness on Home Sale Prices

Notes: Panel A: continuous treatment (effect of 1 SD increase in strictness). Panel B: binary treatment (separate effects for blocks moving to stricter vs. more lenient aldermen). 2015 redistricting cohort. Border-pair-by-side and border-pair-by-year fixed effects. Triangular kernel weights with 1,000 ft bandwidth. Includes hedonic controls for building square footage, land square footage, building age, bedrooms, bathrooms, and garage. Year −1 omitted as reference. 95% confidence intervals based on standard errors clustered at the census block level.

Table 1: Summary Statistics of New Residential Construction

	(1) All New Construction	(2) Multifamily Only
Average FAR	1.22	2.32
Average Units	6.65	31.27
Average Stories	2.27	2.75
Average DUPAC	28.47	74.54
Median Dist. to Boundary (ft)	928.57	887.16
Average Strictness Score	0.154	0.285
Observations	13,236	2,469

Table 2: Effect of Aldermanic Strictness on Development Density

	ln(FAR) (1)	ln(DUPAC) (2)	ln(Units) (3)
Strictness Score	-0.13**	-0.10**	-0.16**
	(0.05)	(0.04)	(0.06)
Observations	334	334	334
Dep. Var. Mean	2.05	63.79	12.16
Ward Pairs	89	89	89
Zoning Code × Ward-Pair FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: Sample includes new multifamily construction (2-100 units) within 250 feet of ward boundaries, 2006-2024. All specifications include border-pair × zoning fixed effects and construction year fixed effects. Strictness is the standardized aldermanic strictness score. Standard errors clustered at the ward-pair level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Summary Statistics: Event Study Samples

	Full Sample	Control	To Stricter	To Lenient
Panel A: Home Sales (2012 Announcement Cohort)				
Transactions	37,891	21,171	8,098	8,622
Census blocks	11,964	6,738	2,498	2,728
Ward boundary pairs	145	125	63	75
Mean sale price ($)	$308,367	$320,524	$259,205	$324,690
Median sale price ($)	$226,000	$228,000	$220,000	$235,000
Mean Δ strictness	—	0.000	0.688	-0.539
Mean dist. to boundary (ft)	487	509	462	454
Mean building sqft	1,942	1,993	1,828	1,923
Mean building age (years)	92.4	92.8	91.3	92.3
Mean bedrooms	4.02	4.09	3.87	3.96
Panel B: Rental Listings (Stacked 2015 + 2023 Cohorts)				
Listings	3,880,857	1,830,997	918,679	1,131,181
Census blocks	13,568	9,544	3,220	3,784
Ward boundary pairs	161	147	97	103
Mean rent ($)	$2,540	$2,551	$2,620	$2,456
Median rent ($)	$2,345	$2,395	$2,365	$2,258
Mean Δ strictness	—	0.000	1.099	-0.484
Mean dist. to boundary (ft)	440	507	318	430
Mean sqft	979	990	968	971
Mean bedrooms	1.58	1.63	1.53	1.54
Share single-family (%)	5.0	5.9	3.1	5.2
Share multi-family (%)	82.5	81.0	84.4	83.3
Share condo (%)	5.5	6.8	3.9	4.5

Notes: Sample restricted to transactions/listings within 1,000 feet of ward boundaries. Panel A uses the 2012 announcement cohort for the 2015 redistricting (sales 2007–2017). Panel B pools the 2015 and 2023 redistricting cohorts using implementation timing; rental sample restricted to listings with non-missing hedonic characteristics. Control units did not switch aldermen during redistricting. Strictness is standardized (mean 0, SD 1) based on permit processing times.

Table 4: Effect of Alderman Strictness on Housing Costs

	Rents		Home Sale Prices	
	(1)	(2)	(3)	(4)
Post × Strictness Δ	0.055***	0.037**	0.022***	0.022***
	(0.020)	(0.015)	(0.009)	(0.007)
Observations	3,877,588	3,877,588	43,895	43,892
R^2	0.40	0.80	0.69	0.74
Hedonic Controls		✓		✓
Border Pair × Side FE	✓	✓	✓	✓
Border Pair × Year FE	✓	✓	✓	✓

Notes: Columns (1)–(2): listing-level regressions of log rent on post-redistricting indicator interacted with change in alderman strictness; stacked estimator combining 2015 and 2023 redistricting cohorts. Columns (3)–(4): transaction-level regressions of log sale price using 2015 implementation timing. Rental hedonic controls include log sqft, log bedrooms, log bathrooms, and building type. Sales hedonic controls include log building sqft, log land sqft, log building age, log bedrooms, log bathrooms, and garage indicator. Sample restricted to observations within 1,000 feet of ward boundaries. Triangular kernel weights. Standard errors clustered by census block in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A Data and Sample Construction

This appendix documents the data sources and sample construction procedures for the analysis.

A.1 Data Sources

The analysis draws on several administrative and proprietary datasets:

- **Cook County Assessor Data**: Sales records, property assessments, and improvement characteristics for residential parcels in Cook County.
- **Chicago Building Permits**: Permit applications and processing records for high-discretion permit types requiring aldermanic approval.
- **RentHub Rental Listings**: Daily snapshots of rental listings in Chicago from 2014–2025.
- **Ward Boundary Shapefiles**: Official ward boundaries for three eras: pre-2015, 2015–2023, and post-2023.
- **Census Data**: American Community Survey block group demographics and 2010/2020 decennial census block boundaries.

A.2 High-Discretion Permits: Sample Construction

The building permits data come from the City of Chicago's open data portal and include all permit applications and processing records.

Sample Restrictions. I restrict the sample to "high-discretion" permit types that are most likely to require aldermanic engagement and face delays due to community meetings or aldermanic intervention. Specifically, I include permits designated as: (1) New Construction; (2) Renovation; (3) Demolition; (4) Porch Construction; and (5) permits that were reinstated after being previously revoked.

I exclude permits that follow expedited review processes: (1) permits in the "Easy Permit Process"; (2) permits in the "Express Permit Program"; (3) signs permits; and (4) scaffolding permits. These exclusions target permit types that bypass standard aldermanic review and thus do not reflect aldermanic influence on development.

Processing Time. For each permit, I compute processing time as the number of days between the application date and the issue date. Permits with missing dates or negative processing times are excluded.

Ward Assignment. Permits are assigned to wards based on the permit's geographic coordinates and the ward boundaries in effect at the time of application.

A.3 Alderman Strictness Scores

Alderman strictness is measured using permit processing times from the high-discretion permit sample described in Section A.2, residualized to remove the influence of ward characteristics.

Stage 1: Residualization. I regress log mean processing time on ward geographic fundamentals (distance to CBD, lakefront share, CTA station density, community area shares), demographic controls (homeownership rate, population, median household income, racial composition), and month fixed effects.

Stage 2: Alderman Fixed Effects. Residualized processing times are regressed on alderman fixed effects, with observations weighted by the number of permits. The alderman fixed effect captures the systematic component of processing time attributable to the alderman.

Empirical Bayes Shrinkage. Raw alderman fixed effects are shrunk toward the grand mean using an empirical Bayes procedure: $\hat{\theta}_j^{\text{EB}} = B_j \cdot \hat{\theta}_j$, where $B_j = \hat{\tau}^2/(\hat{\tau}^2 + \hat{\sigma}_j^2)$ is the shrinkage factor.

Index Standardization. The final strictness index is standardized to have mean zero across aldermen. Higher values indicate stricter aldermen.

Table A.1 reports the Stage 1 residualization regression. The dependent variable is log mean processing time at the ward-month level, weighted by the number of permits. The coefficients reveal systematic relationships between ward characteristics and permit processing times. Wards closer to the CBD and with higher median incomes have shorter processing times, while wards with greater lakefront exposure, more CTA stations, and higher homeownership rates have longer processing times. These patterns are absorbed in the residualization, ensuring that the alderman fixed effects capture only the component of processing time attributable to the alderman rather than ward characteristics.

A.4 New Construction: Sample Construction

The new construction sample for the density analysis is drawn from the Cook County Assessor's parcel-level data.

Sample Restrictions. I apply the following restrictions: (1) lot area exceeds 1 square foot; (2) building area exceeds 1 square foot; (3) at least one dwelling unit; and (4) construction year is 2006 or later. These restrictions exclude vacant parcels, data errors, and older buildings that predate the sample period.

Density Measures. For each parcel, I compute two density measures: Floor Area Ratio (FAR), defined as total building square footage divided by lot area; and Dwelling Units Per Acre (DUPAC), defined as the number of dwelling units divided by lot acreage.

Ward and Boundary Assignment. Each parcel is assigned to a ward based on its centroid location and the ward boundaries in effect at the time of construction. Distance to the nearest ward boundary is computed in feet using the Illinois East State Plane coordinate system.

A.5 Summary Statistics: New Construction Sample

Table 1 presents summary statistics for the new residential construction sample used in the density analysis. The sample includes all new residential buildings constructed in Chicago between 2006 and 2025 that meet the restrictions described in Section A.4. Column (1) reports statistics for the full sample, while Column (2) restricts to multifamily buildings with 2 or more units.

The full sample contains 13,236 new residential buildings with an average of 6.65 dwelling units and 2.27 stories. The average Floor Area Ratio (FAR) is 1.22 and the average Dwelling Units Per Acre (DUPAC) is 28.47. Multifamily buildings are substantially denser, with an average of 31.27 units, 2.75 stories, FAR of 2.32, and DUPAC of 74.54. Both samples have similar median distances to the nearest ward boundary (approximately 900 feet).

A.6 Home Sales: Sample Construction

The home sales data come from the Cook County Assessor's Office and include all residential property transactions in Chicago from 1999–2025.

Sample Restrictions. I apply the following restrictions to construct the analysis sample: (1) residential property classes only, including single-family homes (classes 202–210) and small multifamily buildings with 2–6 units (class 211), excluding mixed-use properties (class 212) and condominiums (classes 299, 399); (2) sale price exceeds $10,000 to exclude nominal transfers and gifts; (3) market transactions only, defined as warranty or trustee deeds; (4) excludes land-only sales; (5) seller and buyer names are valid and non-identical, excluding intrafamily transfers; and (6) single-parcel transactions only.

Price Winsorization. Sale prices are winsorized at the 1st and 99th percentiles to limit the influence of outliers on mean calculations.

Geocoding. Each sale is geocoded by joining the property identification number (PIN) to the Cook County parcel universe, which contains latitude and longitude coordinates. Sales that cannot be matched on the full 14-digit PIN are matched on the first 10 digits (PIN10).

Ward Assignment. Sales are assigned to wards based on the ward boundaries in effect at the exact sale date. The ward map changes on May 18, 2015 and May 15, 2023. For the event study analysis, sales before May 18, 2015 use pre-2015 boundaries; sales between May 18, 2015 and May 15, 2023 use 2015 boundaries; and sales after May 15, 2023 use post-2023 boundaries.

Distance to Ward Boundary. Distance calculations follow the same procedure as for new construction (Section A.4), using the Illinois East State Plane coordinate system (EPSG:3435) and computing signed distance based on relative alderman strictness.

A.7 Rental Listings: Sample Construction

The rental listings data contain daily snapshots of active listings from RentHub. Raw variables include listing price, unit characteristics (bedrooms, bathrooms, year built), building type, amenity indicators (laundry, gym), and geographic coordinates.

Sample Restrictions. I apply the following restrictions to construct the analysis sample: (1) valid rental price (non-missing and positive); (2) valid geographic coordinates (non-missing latitude and longitude); and (3) successfully geocoded to a Chicago ward.

Spatial Assignment. Each listing is assigned to the relevant ward based on its file date and the ward boundaries in effect at that time. The ward map changes on May 18, 2015 (following the 2015 aldermanic elections) and May 15, 2023 (following the 2023 redistricting). Census block assignment uses 2010 blocks for the 2015 cohort and 2020 blocks for the 2023 cohort.

Distance to Ward Boundary. For each listing, I calculate the distance to the nearest ward boundary. Shared ward boundaries are constructed via polygon intersection, and the nearest boundary segment is identified using spatial indexing. Distance is computed in the Illinois East State Plane coordinate system (EPSG:3435) and expressed in feet. The signed distance is positive when the listing is in the stricter alderman's ward and negative otherwise.

Building Type Classification. Raw building type strings are classified as: Single-family (matches SFR, Single-family, house); Multi-family (matches APT, Apartment, Multi-family, duplex, triplex, fourplex); Condo (matches CON, Condo, Condominium); Townhouse (matches TH, Townhouse); and Other (all remaining).

A.8 Rental Listings: Summary Statistics

Table A.2 presents summary statistics for the rental listings sample.

A.9 Event Study Panel Construction

For the event study analysis, I construct block-level panels around the 2015 and 2023 redistricting events.

Treatment Assignment. Blocks are classified as treated if they were redistricted from one ward to another. For the 2015 cohort, I compare ward assignments under the pre-2015 map (2014 boundaries) to post-2015 (2015 boundaries). For the 2023 cohort, I compare 2022 boundaries to 2024 boundaries. Census blocks are assigned to wards based on their centroid location.

Predetermined Strictness. Treatment intensity is measured using predetermined alderman strictness scores—scores computed from permit data in the year before redistricting takes effect. For the 2015 cohort, I use 2014 strictness scores for both origin and destination wards. For the 2023 cohort, I use 2022 scores. This ensures treatment intensity is

not affected by post-treatment changes in aldermanic behavior. The change in strictness for block i is: $\Delta\text{Strictness}_i = \text{Strictness}_{\text{dest}} - \text{Strictness}_{\text{origin}}$. For binary treatment specifications, blocks are classified as "moved to stricter" if $\Delta\text{Strictness}_i > 0$ and "moved to more lenient" if $\Delta\text{Strictness}_i < 0$.

Contaminated Control Exclusion. Control blocks may experience treatment through electoral turnover (alderman changed via election rather than redistricting). I identify wards with alderman turnover between 2014–2015 and 2022–2023 and exclude non-switching blocks in these wards from the control group.

Panel Aggregation. Rental listings and home sales are aggregated separately to block-year and block-quarter levels. For rentals, I compute number of listings, mean rent, and median rent. For sales, I compute number of sales, mean price, and median price. The final panels stack the 2015 and 2023 cohorts with cohort-specific fixed effects. The 2015 cohort includes years 2010–2020, while the 2023 cohort includes years 2018–2025, providing a symmetric five-year window around each redistricting event.

Table A.1: Stage 1 Residualization: Log Mean Processing Time on Ward Fundamentals

	Log Processing Time (1)
Dist. to CBD (km)	0.021*
	(0.011)
Lakefront share (≤1km)	-1.06***
	(0.271)
CTA stations (≤800m)	0.008*
	(0.004)
Homeownership rate	-0.401***
	(0.135)
Population (10,000s)	-0.090***
	(0.018)
Median HH income ($10,000s)	-0.038***
	(0.010)
Share Black	-0.649***
	(0.125)
Share Hispanic	-0.490***
	(0.120)
Observations	10,646
R^2	0.43260
Community Area FE	✓
Month FE	✓

Notes: Dependent variable is log mean permit processing time at the ward-month level. Observations weighted by the number of high-discretion permits. Community area fixed effects (77 areas) included but not shown. Month fixed effects absorb city-wide temporal variation in processing times.

Table A.2: Rental Listings Summary Statistics

	Mean	SD	P10	P50	P90	N
Panel A: Price and Unit Characteristics						
Rent ($/month)	2,210	1,152	1,115	1,995	3,443	9,940,956
Bedrooms	1.37	0.99	0.00	1.00	3.00	9,940,956
Bathrooms	1.29	0.52	1.00	1.00	2.00	9,927,976
Year built	1994	29	1951	2009	2016	347,026
Panel B: Building Type Distribution	Count			Percent		
Multi-family	7,319,452			73.6%		
Single-family	1,030,517			10.4%		
Condo	810,699			8.2%		
Other	722,549			7.3%		
Townhouse	57,739			0.6%		
Panel C: Amenities			Percent Present			N
Laundry			59.3%			9,940,956
Gym			42.5%			9,940,956
Panel D: Location						
Distance to boundary (ft)	1,222	1,218	139	851	2,785	9,940,956

Notes: Sample includes 9,940,956 rental listings from 2014–2025 with valid price, coordinates, and ward assignment. Distance to boundary is the absolute distance to the nearest shared ward boundary.

B Additional Density Results

B.1 Bandwidth Robustness: Border-Pair Fixed Effects

Table B.1 presents border-pair fixed effects estimates using a 1,000 foot bandwidth, compared to the 250 foot bandwidth in the main text. The coefficients are smaller in magnitude and less precisely estimated, consistent with the interpretation that aldermanic effects are most pronounced very close to ward boundaries where treated and control parcels are most comparable.

B.2 Covariate Balance Test

A key identifying assumption in the border-pair fixed effects design is that neighborhood characteristics vary smoothly across ward boundaries, so any discontinuity in development outcomes can be attributed to differences in aldermanic behavior rather than underlying area characteristics.

To test this assumption, I assign each parcel to a census block group via spatial intersection with 2019 American Community Survey (ACS) geographies. Block groups are small geographic units (typically 600–3,000 residents) that provide demographic variation at a much finer scale than wards. I then merge block group-level demographics from the five-year ACS estimates corresponding to each parcel's construction year (using 2010–2014 estimates for pre-2015 construction and 2015–2019 estimates for later construction).

I regress each demographic variable on the aldermanic strictness score using the same fixed effects specification as my main regressions (zone code $\times$ ward-pair and year fixed effects), restricting to parcels within 250 feet of the border.

B.3 Donut Robustness: Border-Pair Fixed Effects

As an additional robustness check for the border-pair fixed effects results, I exclude parcels within 25 or 50 feet of the ward boundary. This addresses concerns that measurement error or other idiosyncratic features of parcels located precisely at the border may be unusual in ways that drive the results.

B.4 RD Robustness Figures

B.4.1 Donut Hole Specification

One concern with spatial RD designs is that observations precisely at the boundary may be unusual in ways that drive the results—for example, if ward boundaries tend to follow major streets or other geographic features that independently affect development patterns. To address this, I implement a "donut RD" (Barreca et al., 2011, 2016) specification that excludes parcels within 25 or 50 feet of the ward boundary. Figure B.1 shows that the discontinuity persists even when dropping observations closest to the cutoff, suggesting the results are not driven by idiosyncratic features of parcels located precisely on ward boundaries.

B.4.2 Functional Form

The baseline RD specification uses a local linear regression. However, results could be sensitive to this functional form assumption, particularly if the underlying relationship between distance and density is nonlinear. Figure B.2 presents estimates using linear, quadratic, and cubic polynomial specifications. The discontinuity is evident under all three specifications. Notably, the higher-order polynomials yield larger point estimates, suggesting that if anything, the linear specification provides a conservative estimate of the treatment effect.

B.4.3 Placebo Boundary Tests

A key identifying assumption of the RD design is that the discontinuity occurs precisely at the ward boundary and not at other arbitrary distances. To test this, I conduct placebo tests by artificially shifting the "cutoff" to locations 500 feet inside each ward (both toward the stricter and more lenient sides). If the discontinuity is truly driven by the ward boundary rather than some other spatial pattern, one should observe no discontinuity at these placebo cutoffs. Figure B.3 confirms this prediction: the placebo boundaries show no significant discontinuity, while the true boundary (at zero) exhibits the large drop documented in the main results.

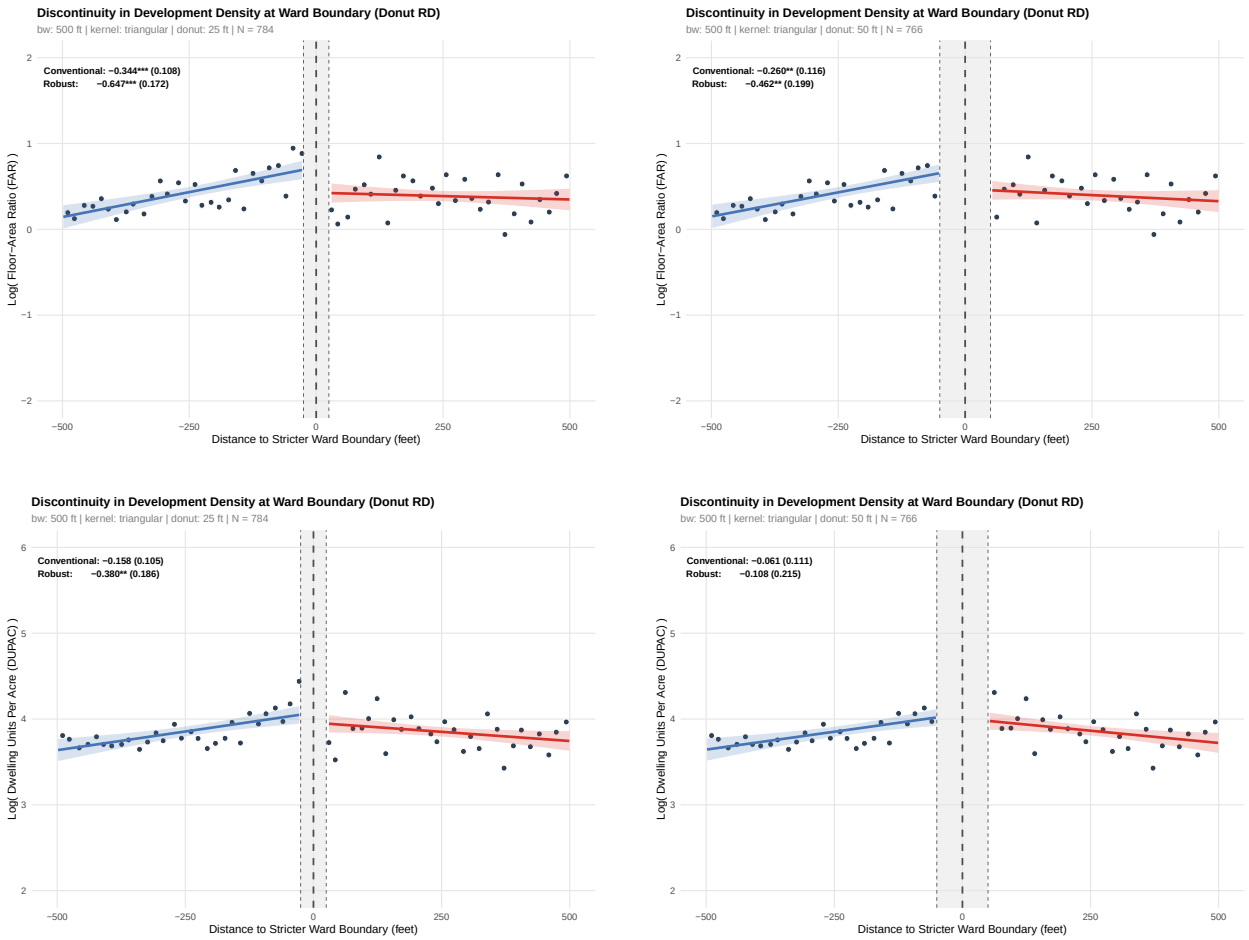


Figure B.1: Donut RD Robustness

Notes: Top row: Log(FAR); bottom row: Log(DUPAC). Left column excludes parcels within 25 feet of boundary; right column excludes parcels within 50 feet. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

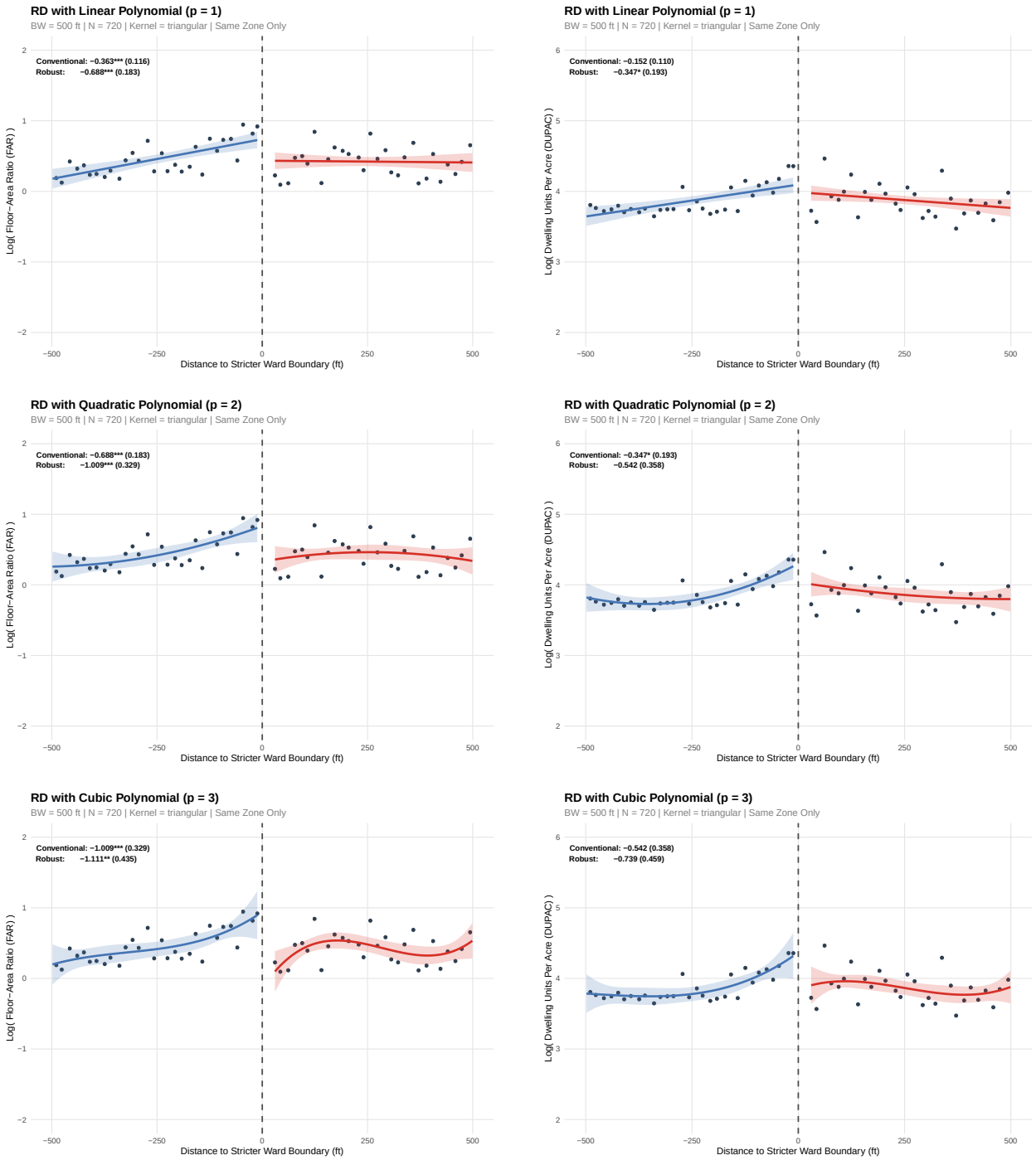


Figure B.2: Functional Form Robustness

Notes: Left column: Log(FAR); right column: Log(DUPAC). Top row: linear ($p = 1$); middle row: quadratic ($p = 2$); bottom row: cubic ($p = 3$). All specifications use 500-foot bandwidth with triangular kernel. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

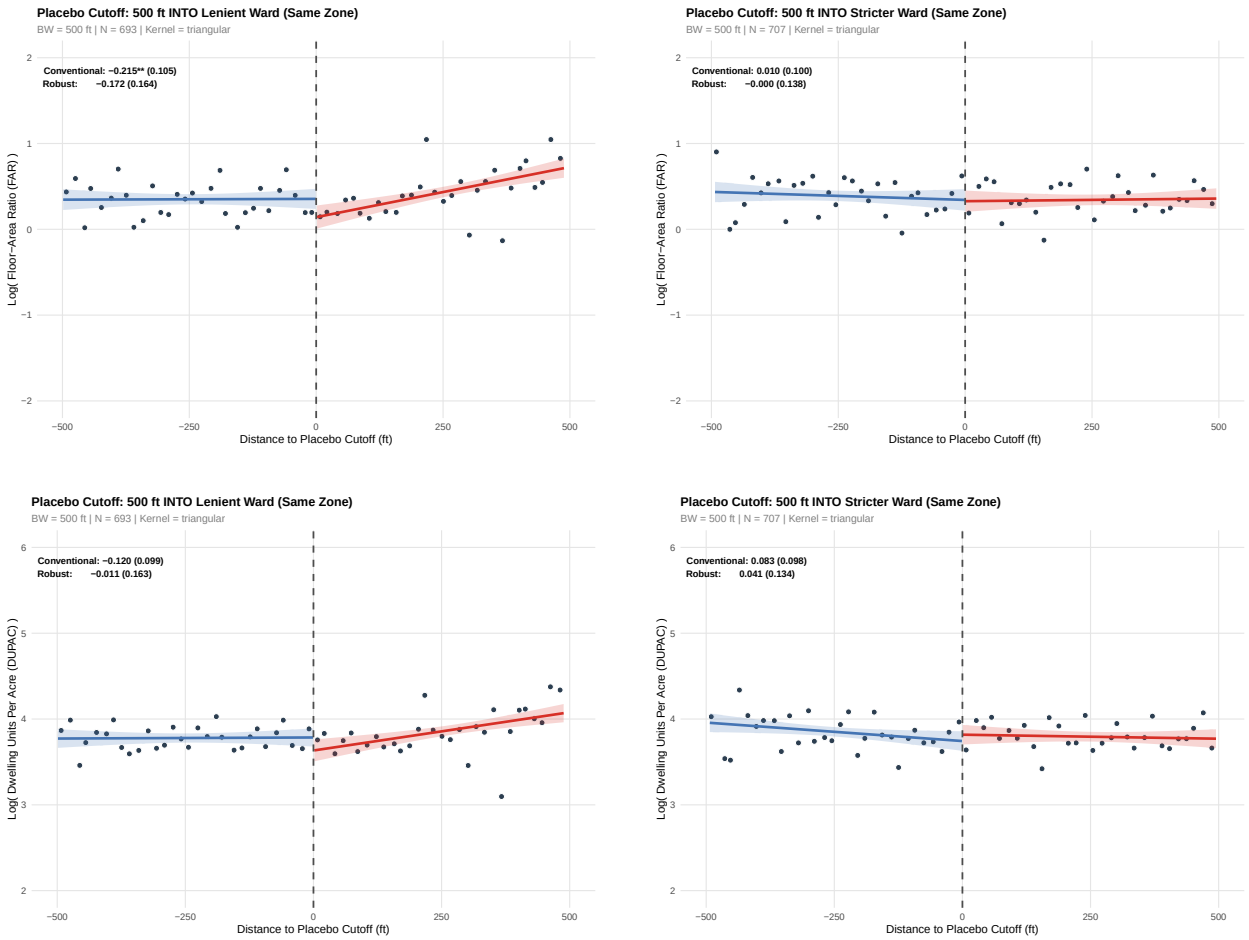


Figure B.3: Placebo Boundary Tests

Notes: Top row: Log(FAR); bottom row: Log(DUPAC). Left column: placebo cutoff at −500 feet (inside lenient ward); right column: placebo cutoff at +500 feet (inside strict ward). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.1: Effect of Aldermanic Strictness on Development Density (1,000 ft Bandwidth)

	ln(FAR) (1)	ln(DUPAC) (2)	ln(Units) (3)
Strictness Score	-0.02	-0.05	-0.07**
	(0.04)	(0.04)	(0.03)
Observations	929	929	929
Dep. Var. Mean	1.82	55.33	9.83
Ward Pairs	118	118	118
Zoning Code × Ward-Pair FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: Sample includes new multifamily construction (2–100 units) within 1,000 feet of ward boundaries, 2006–2024. All specifications include border-pair × zoning fixed effects and construction year fixed effects. Standard errors clustered at the ward-pair level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: Covariate Balance Test: Block Group Demographics

Panel A: Race, Income, and Education

	% White (1)	% Black (2)	Median Income (3)	% Bachelor's+ (4)
Strictness Score	-0.002	0.029	-7,306.6*	-0.020
	(0.022)	(0.023)	(3,736.9)	(0.020)
Observations	367	367	359	367
Dep. Var. Mean	0.36	0.30	68008.57	0.44
Ward Pairs	89	89	89	89
Zoning Code × Ward-Pair FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Panel B: Housing and Density

	Homeownership (1)	Median Rent (2)	Home Value (3)	Pop. Density (4)
Strictness Score	-0.031**	-59.2	2,749.1	-623.4
	(0.015)	(44.7)	(8,063.3)	(402.0)
Observations	367	363	350	367
Dep. Var. Mean	0.40	1243.94	333277.23	7763.78
Ward Pairs	89	89	89	89
Zoning Code × Ward-Pair FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: Each column regresses a census block group demographic variable on the aldermanic strictness score using the same specification as Table 1 (border-pair × zoning and year fixed effects). Sample: parcels within 250 feet of ward boundaries. Demographics from 2019 ACS 5-year estimates. Null results indicate strictness is uncorrelated with observable neighborhood characteristics at the boundary. Standard errors clustered at the ward-pair level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.3: Donut Robustness: Excluding Parcels Within 25 Feet of Boundary

	ln(FAR) (1)	ln(DUPAC) (2)	ln(Units) (3)
Strictness Score	-0.13**	-0.12**	-0.15**
	(0.06)	(0.05)	(0.06)
Observations	331	331	331
Dep. Var. Mean	2.05	63.71	12.21
Ward Pairs	89	89	89
Zoning Code × Ward-Pair FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: Same specification as Table 1 but excluding parcels within 25 feet of the ward boundary. Sample: new multifamily construction (2–100 units), 25–250 feet from boundary. Standard errors clustered at the ward-pair level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.4: Donut Robustness: Excluding Parcels Within 50 Feet of Boundary

	ln(FAR) (1)	ln(DUPAC) (2)	ln(Units) (3)
Strictness Score	-0.13**	-0.12**	-0.15**
	(0.05)	(0.05)	(0.06)
Observations	317	317	317
Dep. Var. Mean	2.06	64.13	12.46
Ward Pairs	88	88	88
Zoning Code × Ward-Pair FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: Same specification as Table 1 but excluding parcels within 50 feet of the ward boundary. Sample: new multifamily construction (2–100 units), 50–250 feet from boundary. Standard errors clustered at the ward-pair level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Additional Home Sales and Rental Results

This appendix presents additional results for home sale prices and rents, complementing the event study analysis in Section 5.

C.1 Event Study Without Hedonic Controls

The main text presents event study results including property-level hedonic controls to isolate the pure price effect holding quality constant. This subsection presents results without hedonic controls to capture the total effect of aldermanic strictness on housing costs, including any composition effects operating through changes in the housing stock. Point estimates are generally similar or slightly larger without hedonic controls, suggesting that composition effects account for only a modest portion of the total price response.

C.1.1 Rents Without Hedonic Controls

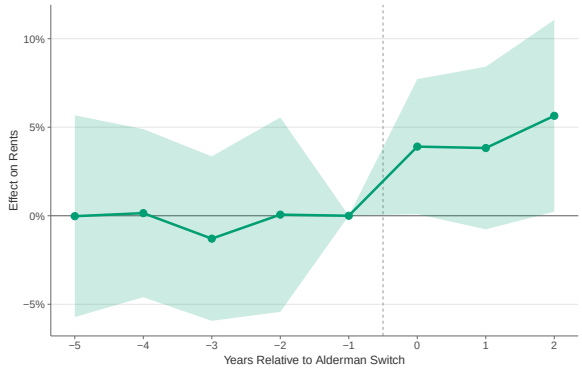
Figure C.1 presents event study results for rents without hedonic controls. The pattern is qualitatively similar to the main specification: flat pre-trends followed by a clear break at redistricting. Point estimates are slightly larger, consistent with part of the effect operating through changes in housing stock composition.

C.1.2 Home Sale Prices Without Hedonic Controls

Figure C.2 presents event study results for home sales without hedonic controls for building square footage, land square footage, building age, number of bedrooms, number of bathrooms, and garage presence. The pattern is qualitatively similar to the main specification with hedonic controls.

C.2 Alternative Event Timing: Announcement vs. Implementation

Table C.1 presents difference-in-differences estimates for home sales using 2012 announcement timing rather than 2015 implementation timing. If housing markets are forward-looking, prices may begin responding when redistricting maps are announced. The point estimates are slightly larger with announcement timing, though the difference is not statistically significant.



Panel A: Continuous Treatment

../tasks/run_event_study_rental_disaggregate/out

Panel B: Binary Treatment

Figure C.1: Event Study: Rents — Without Hedonic Controls

Notes: Same specification as Figure 7 but excluding listing-level hedonic controls. Stacked estimator combining 2015 and 2023 redistricting cohorts. Cohort-by-border-pair-by-side and cohort-by-border-pair-by-year fixed effects. Triangular kernel weights with 1,000 ft bandwidth. Year -1 omitted as reference. Post-period truncated at $t = 2$ where both cohorts contribute observations. 95% confidence intervals based on standard errors clustered at the census block level.

../tasks/run_event_study_sales_disaggregate/outputs/

Panel A: Continuous Treatment

Panel B: Binary Treatment

Figure C.2: Event Study: Home Sale Prices — Without Hedonic Controls

Notes: Same specification as Figure 8 but excluding property-level hedonic controls. 2015 redistricting cohort. Border-pair-by-side and border-pair-by-year fixed effects. Triangular kernel weights with 1,000 ft bandwidth. Year −1 omitted as reference. 95% confidence intervals based on standard errors clustered at the census block level.

Table C.1: Effect of Alderman Strictness on Home Sale Prices: Announcement Timing

	(1)	(2)
Post × Strictness Δ	0.028***	0.028***
	(0.011)	(0.009)
Log Building Sqft		0.349***
		(0.015)
Log Land Sqft		0.235***
		(0.017)
Log Building Age		-0.170***
		(0.006)
Log Bedrooms		-0.046***
		(0.015)
Log Bathrooms		0.043***
		(0.012)
Has Garage		0.017**
		(0.008)
Observations	37,876	37,876
R^2	0.67	0.71

Notes: Transaction-level regressions of log sale price on post-redistricting indicator interacted with change in alderman strictness. Uses 2012 announcement timing as robustness check (main results use 2015 implementation timing). All specifications include border-pair × origin-side and border-pair × year fixed effects. Sample restricted to within 1,000 feet of ward boundaries. Triangular kernel weighting. Standard errors clustered by census block in parentheses. Column (1) excludes hedonic controls; column (2) includes hedonic controls. *** p<0.01, ** p<0.05, * p<0.1.